Differential Diagnosis of Spinal conditions

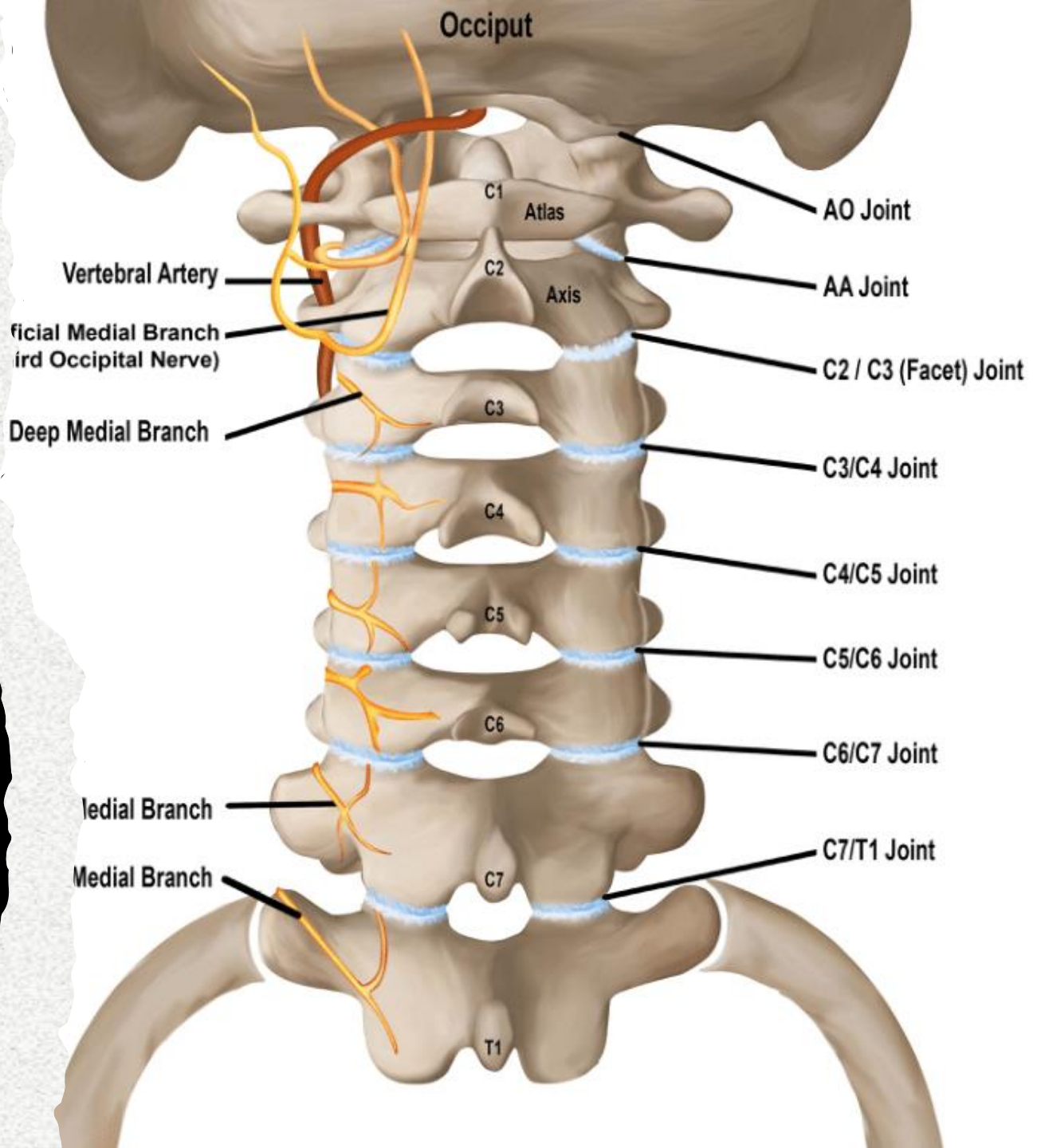
Dr Walaa Abu-Taleb



Aetiology

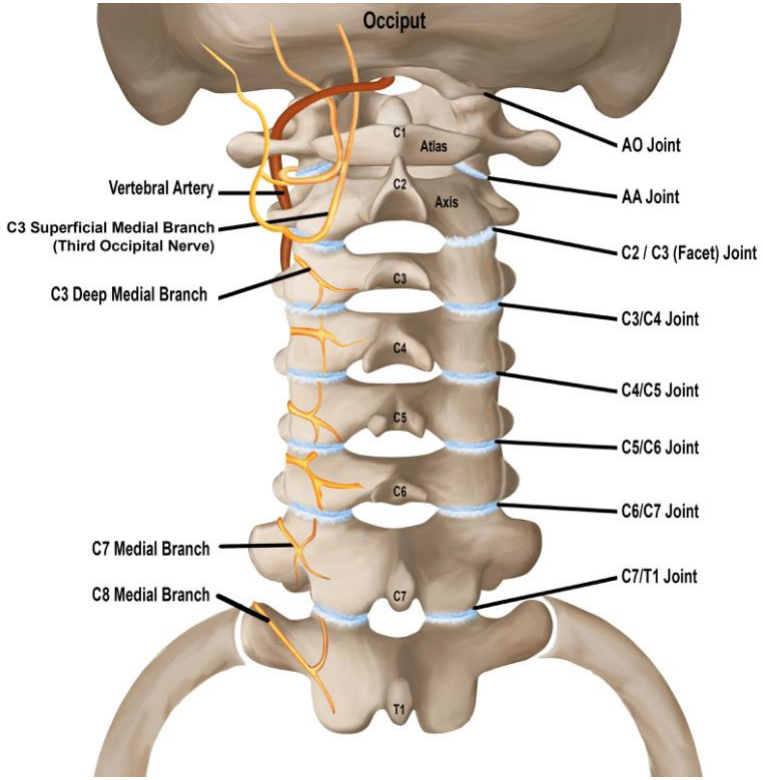
- Various spinal structures including ligaments, facet joints, paravertebral musculature and fascia, intervertebral discs, and spinal nerve roots have been implicated as pain generators.
- The aetiologies can be subdivided into 3 groups: mechanical, systemic, and referred.
- By far, the most common cause is mechanical (97%).

Cervical Spine



Functional Anatomy of cervical spine

Facet capsules, discs, nerve roots, dura, ligaments, and deep neck flexors are the major pain generators.



Upper Cervical (C0–C2)	Rotation at C1–C2 (≈ 50% of total rotation). C0–C1 flexion-extension.	Dysfunctions here commonly cause headaches, dizziness, and non-dermatomal symptoms.
Lower Cervical (C3–C7)	Typical vertebrae, coupling of side bend + rotation to same side.	This is where facet, disc, and radiculopathy patterns show up.

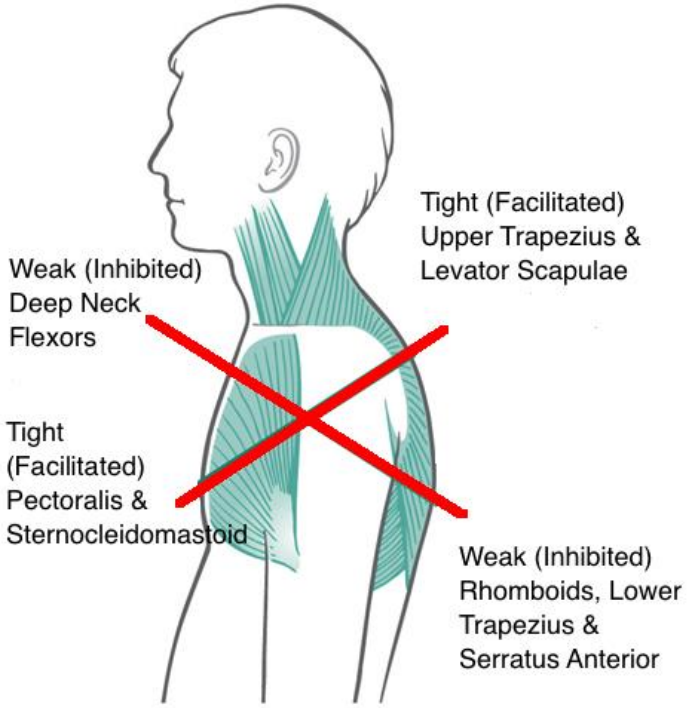
CERVICAL SPINE
DIFFERENTIAL DIAGNOSIS BY ETIOLOGY

A. Mechanical / Postural / Movement Dysfunction

Cervical Postural Dysfunction	Dull ache, worsens with prolonged sitting, improves with movement. No neurological signs.
Facet Joint Syndrome (Zygapophyseal Joint Pain)	Unilateral neck pain, worse with extension + rotation, local tenderness. Pain may refer into shoulder/scapula but not below elbow.
Myofascial Pain Syndrome	Local muscle painful spot within a taut bands (common in UT, levator scapulae), trigger points reproduce symptoms.

Cervical Postural Dysfunction (Forward Head Posture- FHP)

- Cervical postural dysfunction refers to pain and movement impairment that develops because the cervical spine is held in sustained, mechanically inefficient positions for long periods.
- The most common pattern is forward head posture combined with thoracic kyphosis and scapular protraction/elevation.
- It is a problem of muscle imbalance, load intolerance, and sensory-motor control.
- When the head sits forward relative to the body, the deep cervical flexors (longus colli, longus capitis) are underused and progressively weaken. In compensation, the superficial cervical flexors (sternocleidomastoid and scalenes) become overactive.
- At the same time, the suboccipital muscles at the base of the skull shorten to hold the head upright. The upper trapezius and levator scapulae also become overloaded because they must continually help support the weight of the head.
- This sustained imbalance leads to: Increased mechanical strain on the facet joint capsules and posterior cervical ligaments. Elevated resting tone and sensitivity in the upper trapezius and suboccipitals.
- Loss of normal segmental control and smooth cervical motion. The pain therefore comes from fatigue and sensitization, not from acute tissue damage.



FHP symptoms and treatment

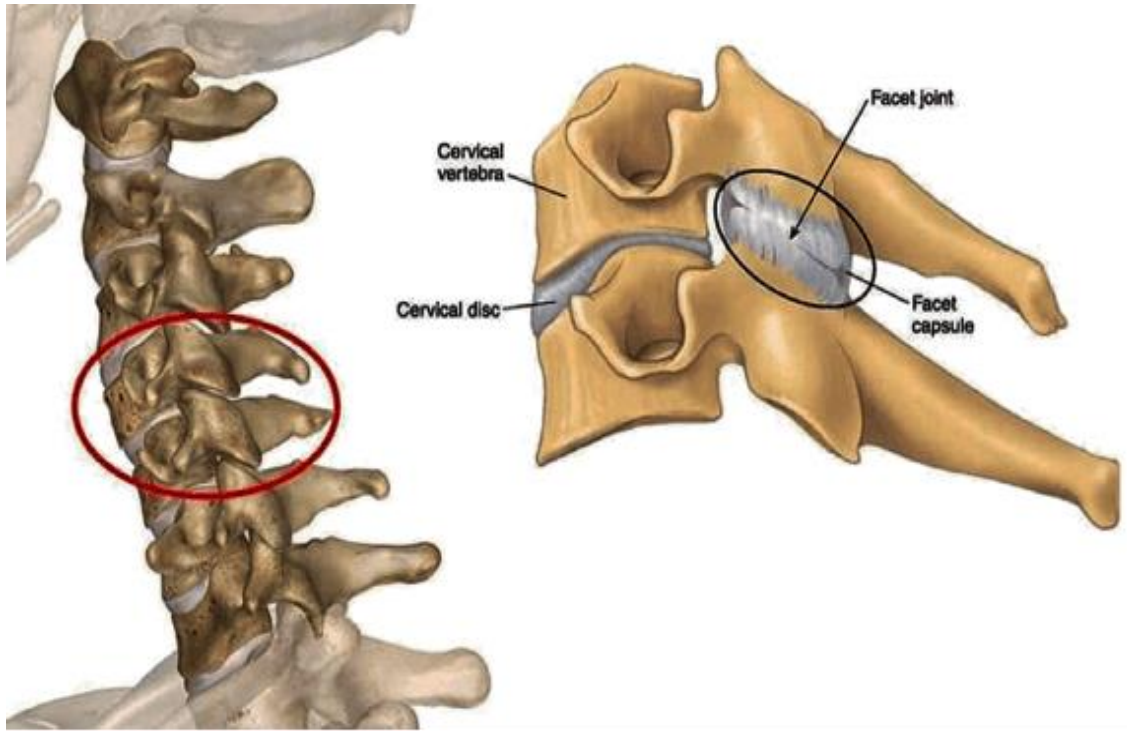
- The pain is usually described as dull, aching, or tight, rather than sharp. The discomfort usually starts in the neck or upper trapezius and may spread toward the scapular region. Headaches may develop when the suboccipital region becomes overloaded.
- Symptoms generally improve with movement or positional change but quickly return when the original sustained posture resumes.
- Symptoms are not dermatomal, and there is no true neurological weakness, numbness, or reflex loss.

Treatment

- Deep neck flexor training sequence (CCF → endurance → functional integration).
- Suboccipital release and postural unloading strategies
- Scapular stabilization sequencing (serratus anterior and lower trapezius emphasis).
- Thoracic mobility and breathing correction.

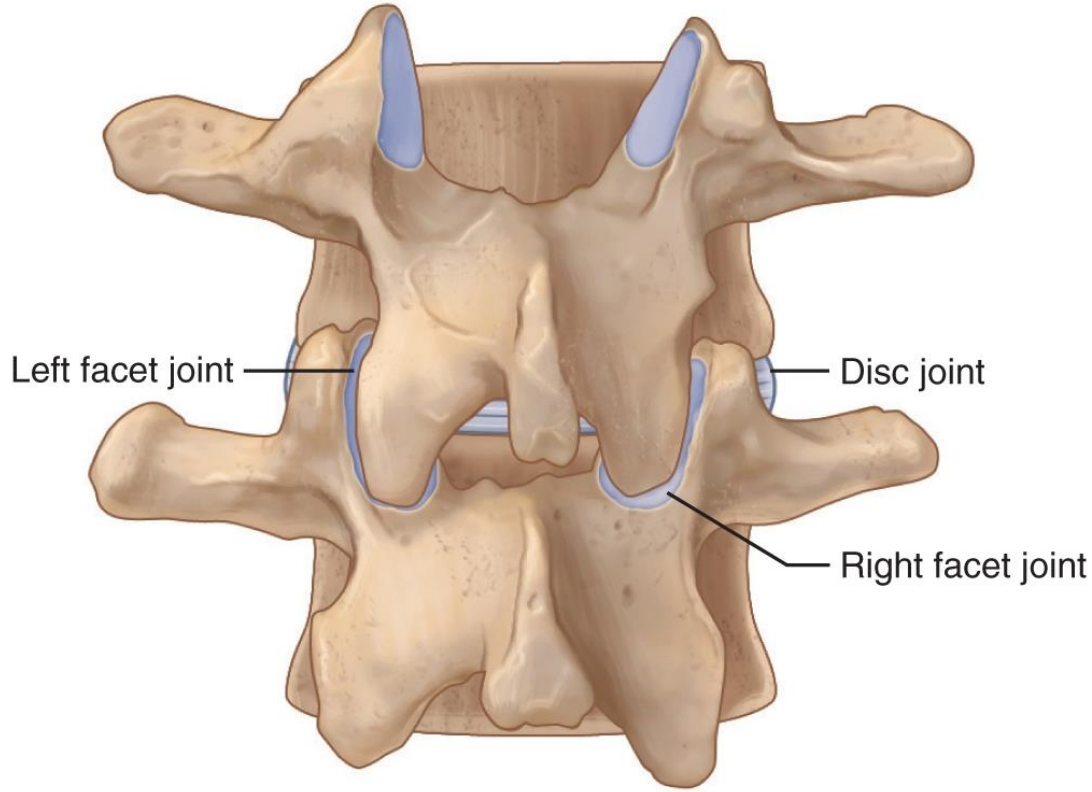
Cervical Facet (Zygapophyseal) Joint Syndrome

- The cervical facet joints are small synovial joints that guide and control movement, especially extension and rotation. Each joint is surrounded by a fibrous capsule that contains pain-sensitive nerve endings.
- When overloaded or irritated, the capsule and the small meniscoid folds inside the joint become inflamed.
- The presence of facet joint changes does not necessarily indicate an origin of pain; similarly, joints that look relatively normal may be the source of symptoms.
- Systems are closely interrelated, and there is association between muscle atrophy and facet joint disease; it may be that spine instability is exacerbated by this connection.



Cervical Facet (Zygapophyseal) Joint Syndrome

- As for all synovial joints, articular cartilage wear and mechanical instability lead to osteoarthritis with localized pain and osteophytes.
- New bone formation around the joint may considerably limit movement which in turn places additional load on adjacent segments.
- As degeneration is inevitably widespread and extremely common, it may be very difficult to determine the origin of pain as the presence of facet joint arthropathy does not predict that pain arises from this area.
- Synovial cysts may arise from the joint and may compress nerve roots. Haemorrhage into the cysts may increase the pressure.



Causes and presentation

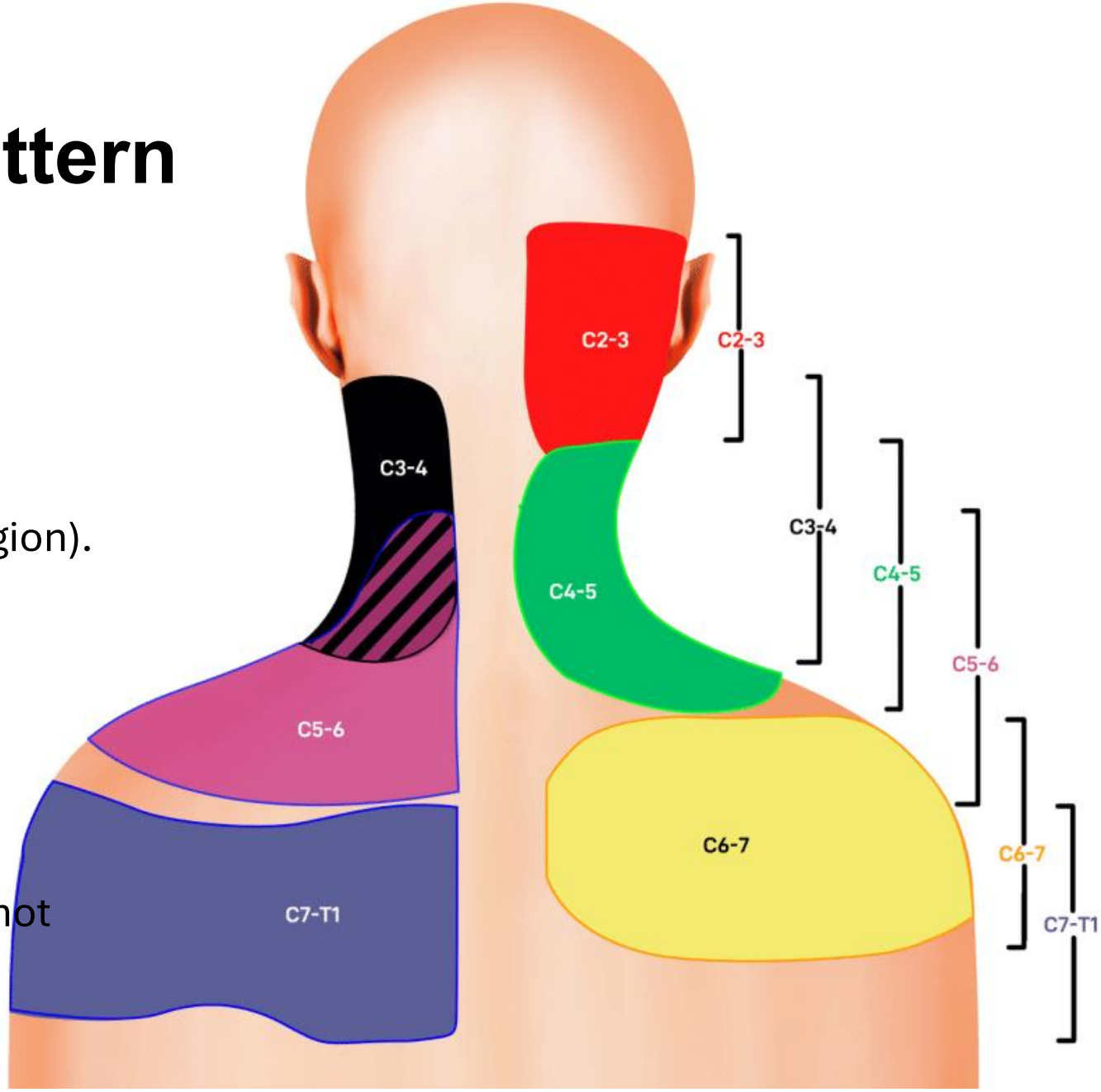
Cervical facet pain typically results from:

- Repeated or sustained extension of the cervical spine
- Rotation and side bending, especially to one side
- Prolonged forward head posture, which causes upper cervical extension

- Sudden “locking” after a quick movement or sleeping awkwardly is reported by patients.
- The pain is usually sharp unilateral and localized to the posterolateral neck. Because the capsule is rich in sensory fibers, even small capsular strains can cause sharp, localized pain.
- It can refer to the upper trapezius, medial border of the scapula, or posterior shoulder.
- Facet pain does not radiate below the elbow.

Facets Referred Pain Pattern

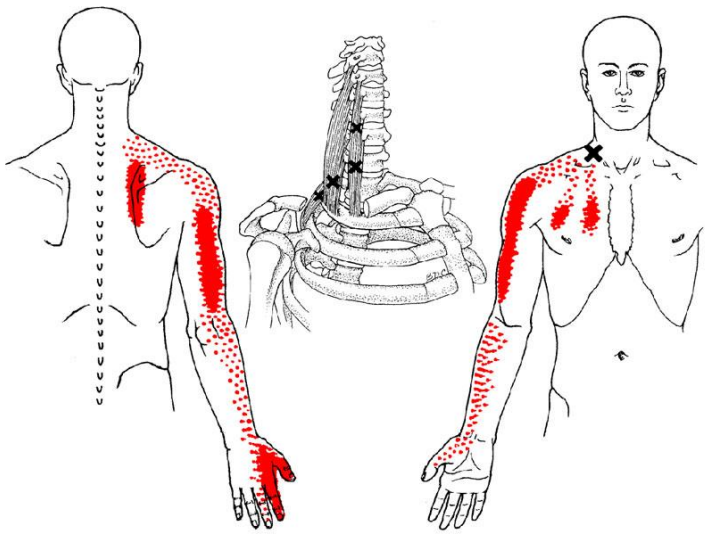
Facets Level	Referred Pain Pattern
C2–3	Occipital headaches (suboccipital region).
C3–4	Upper neck + trapezius.
C4–5	Mid-cervical to scapular region.
C5–6	Interscapular region and posterior shoulder.
C6–7	Scapular border + posterior arm (but not past the elbow).

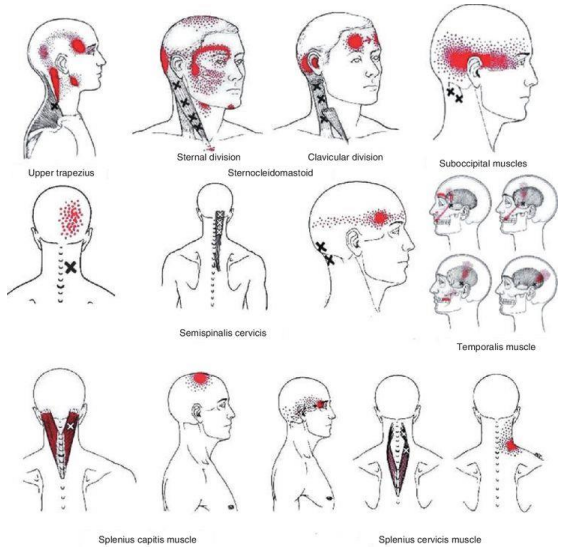


Consensus practice guidelines on interventions for cervical spine (facet) joint pain from a multispecialty international working group

Cervical Trigger Points

- A Trigger Point (TrP) is a hyperirritable spot, a palpable nodule in the taut bands of the skeletal muscles' fascia. Direct compression or muscle contraction can elicit jump sign, local tenderness, local twitch response and referred pain which usually responds with a pain pattern distant from the spot.
- **Jump sign** is the characteristic behavioural response to pressure on a TrP. Individuals are frequently startled by the intense pain. They wince or cry out with a response seemingly out of proportion to the amount of pressure exerted by the examining fingers. They move involuntarily, jerking the shoulder, head, or some other part of the body not being palpated. A jump sign thus reflects the extreme tenderness of a TrP. This sign has been considered pathognomonic for the presence of TrPs.
- **Local twitch response** - defined as a transient visible or palpable contraction of the muscle and skin as the tense muscle fibres contract when pressure is applied through needle penetration or by transverse snapping palpation. A local twitch response on stimulating active TrPs is a widely accepted diagnostic sign.
- **Referred pain**, also called reflective pain, is pain perceived at a location other than the site of the painful stimulus. Pain is reproducible and does not follow dermatomes, myotomes, or nerve roots. There is no specific joint swelling or neurological deficits. Pain from a myofascial TrP is a distinct, discrete and constant pattern or map of pain with no gender or racial differences able to reproduce symptoms - **referred pain map**.



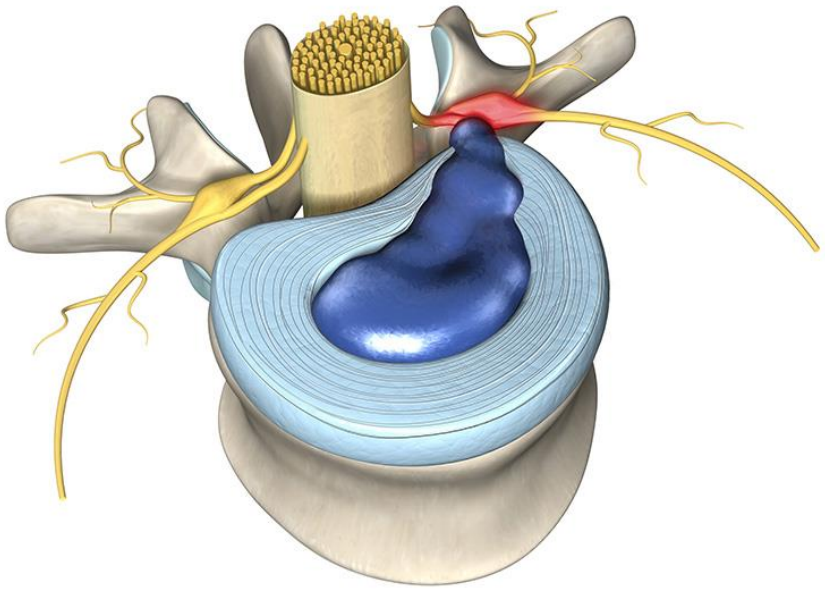


Diagnosis of MTrPs

- **Essential diagnostic criteria:**
    1. Hypersensitive spot
    2. Taut band
    3. Recognised referred pain pattern

- **Supporting diagnostic criteria:**
    1. Local twitch response (LTR)
    2. Reduced range of motion
    3. Reproduction of symptoms

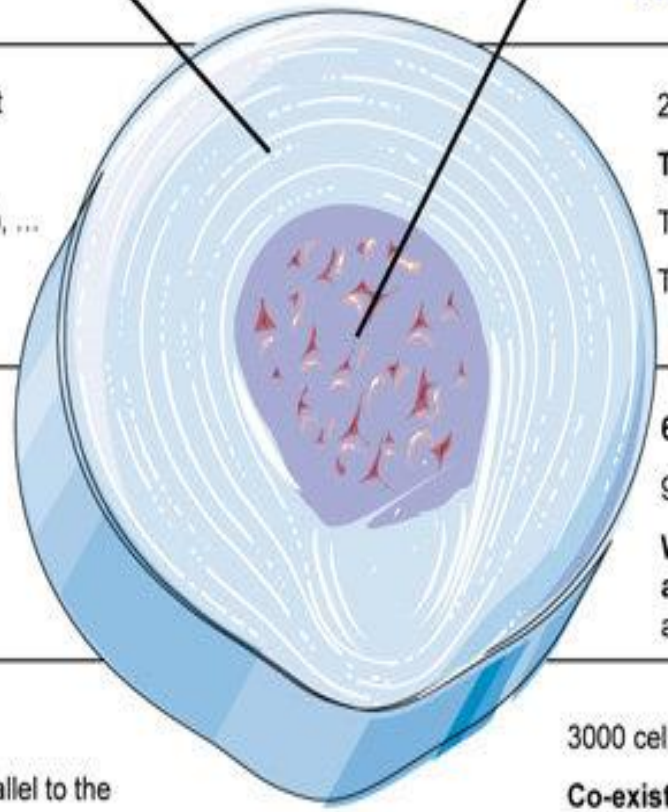
CERVICAL SPINE DIFFERENTIAL DIAGNOSIS BY ETIOLOGY



B. Neurological and Discogenic Conditions

Cervical Disc Herniation	Neck pain + arm pain, worse with flexion, coughing or Valsalva.
Cervical Radiculopathy	Dermatomal pain, positive Spurling, weakness in myotome, reduced reflexes.
Cervical Myelopathy (spinal cord compression – RED FLAG)	Bilateral symptoms, gait disturbance, hand clumsiness, hyperreflexia, positive Babinski/Clonus.

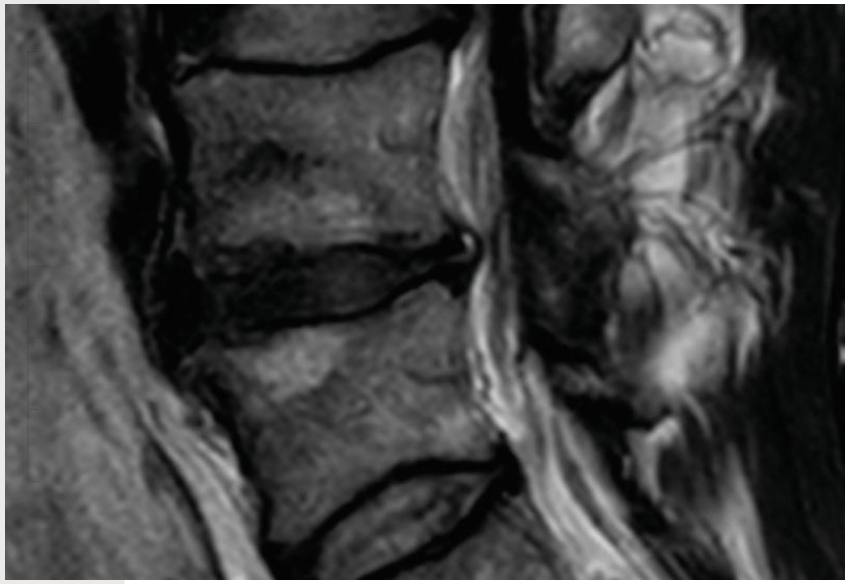
Components of an intervertebral disc

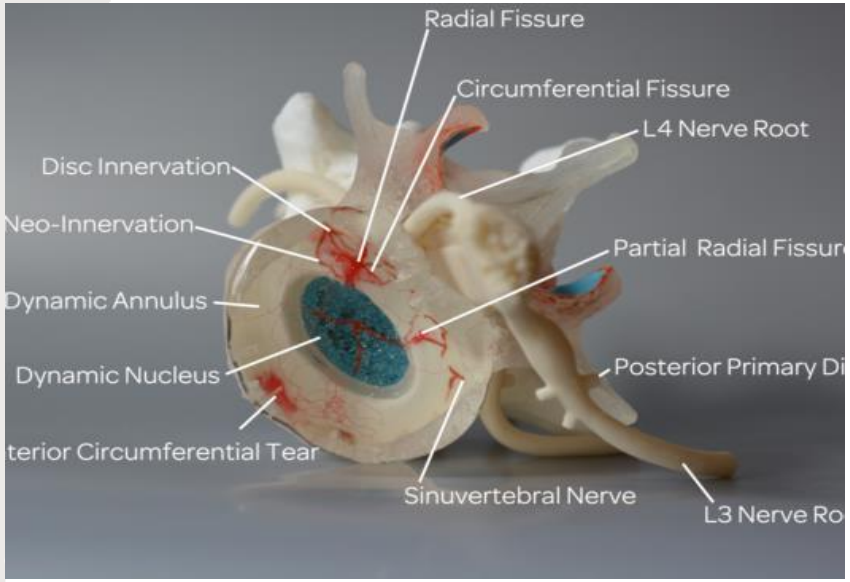


	Annulus fibrosus	*Nucleus pulposus*
Collagen network	65 to 70% of AF dry weight **Type I** collagen (50%), and Type II collagen (30%), ... 7 to 15 concentric bands. Figure-eight arrangement	20% of NP dry weight **Type II** collagen (50%), Type I (20%), ... Tangled fibers
Proteoglycans and water content	30% of AF dry weight 60 to 70% water **No change** during life	**60%** of NP dry weight 90% water (childhood). **Water content decreases with age** (young adult, 80%; older adult, 70%)
Cells	9000 cells/mm^3 Elongated cells lying parallel to the collagen "**Fibroblastic**" phenotype	3000 cells/mm^3 **Co-existence** of two cells types until skeletal maturity is achieved : **notochordal cells** and cells exhibiting a "**chondrocytic**" phenotype

Discs degeneration

- Small tears within the disc lead to annular tears. A tear may be seen as a defect in the annulus. There is considerable debate as to whether these tears are a cause of localized back pain.
- The consequence of all these changes is reduction in pliability and tension within the annulus and a tendency for disc height reduction. This in turn leads to buckling of the longitudinal ligaments.
- Much more important factor in morbidity is mechanical instability due to loss of tension within the annulus and laxity of the margins of the disc. Frank herniation of disc also releases several chemical irritants including prostaglandin, tumour necrosis factor alpha (TNFα) and interleukins.
- These chemicals may be part of the localized pain response and are potentially toxic to the adjacent nerves.
- The loss of height in an intervertebral disc will in turn create mechanical instability, and this will lead to osteophytes that may impinge on adjacent nerves.

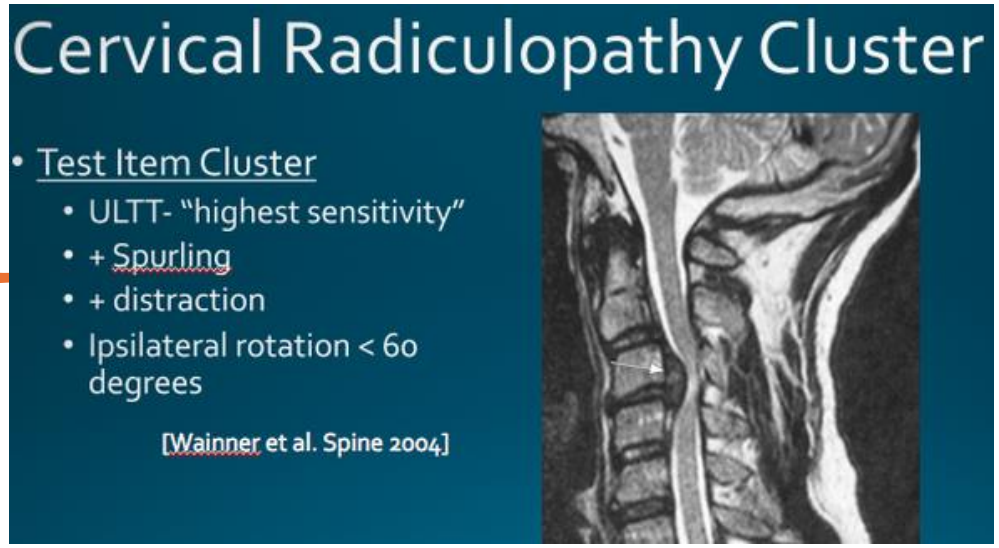




Intervertebral Radicular Pain

Radiculopathy may include dermatomal pain, myotomal weakness , ↓ reflex

Nerve Root	Pain / Sensory	Weakness	Reflex
C5	Lateral shoulder	Deltoid	Biceps
C6	Thumb & radial forearm	Wrist extensors / Biceps	Brachioradialis
C7	Middle finger	Triceps / Wrist flexors	Triceps
C8	4th–5th digits / ulnar hand	Finger flexors	None reliably



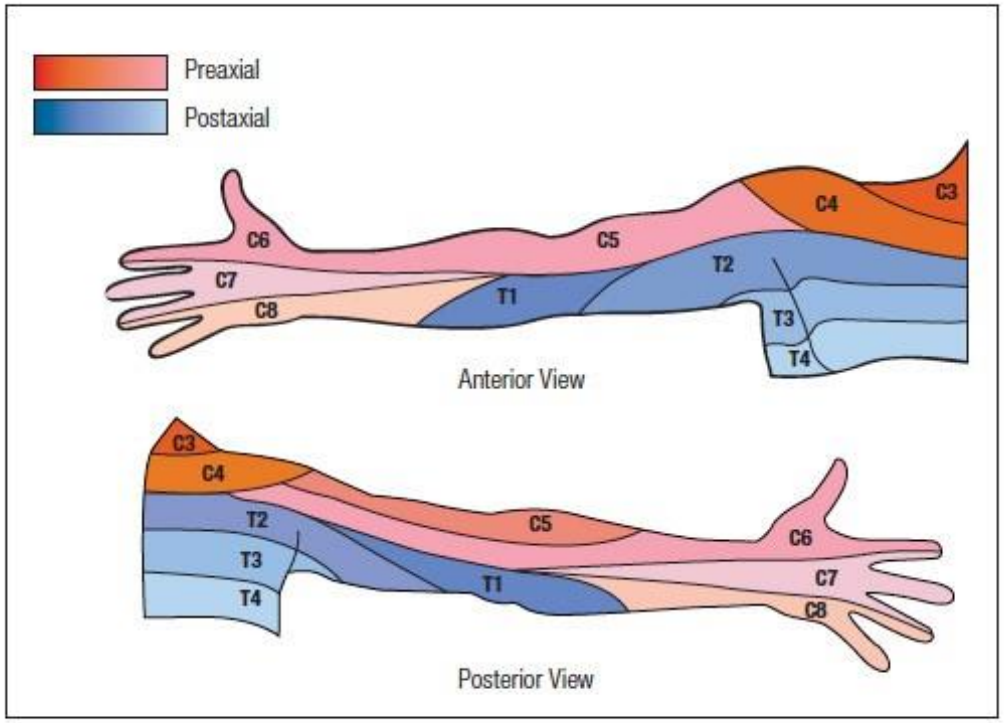


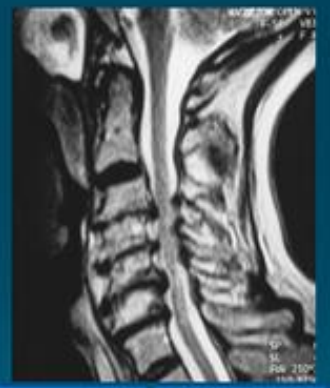
Figure 2. A map of dermatomal innervation of the shoulder and upper extremity.

Cervical myelopathy

- Myelopathy (Spinal Cord Compression – Red flag)
- Bilateral symptoms
- Hand clumsiness (difficulty with buttons / writing)
- Gait imbalance
- Hyperreflexia, Babinski, clonus
- Myelopathy ≠ radiculopathy. Myelopathy = central lesion → affects both sides and legs.

Cervical Myelopathy Cluster

- Age > 45 years
- Gait Ataxia
- Babinski
- Hoffman's Reflex
- Inverted Supinator Sign



Clustered Results	Sensitivity (95% CI)	Specificity (95% CI)	Likelihood Ratio (95% CI) +	Likelihood Ratio (95% CI) -	Posttest Prob of CSM (%)
1 of 5 positive tests	.94 (.89-.97)	.31 (.27-.32)	1.4 (1.2-1.4)	0.18 (0.12-0.42)	43
2 of 5 positive tests	.39 (.33-.46)	.88 (.84-.92)	3.3 (2.1-5.5)	0.63 (0.59-0.79)	64
3 of 5 positive tests	.19 (.15-.20)	.99 (.97-.99)	30.9 (5.5-181.8)	0.81 (0.79-0.87)	94
4 of 5 positive tests	.09 (.06-.09)	1.0 (.98-1.0)	Inf (3.9-Inf)	0.91 (0.90-0.95)	99+



C- Degenerative Conditions: Cervical Spondylosis



Degenerative spine conditions are a group of disorders that cause the spine's gradual loss of normal structure and function, often due to aging, but also from causes like arthritis, infections, or tumors.
Age-related changes, stiffness, morning pain improves with activity.

Pathophysiology
Cumulative mechanical loading and ageing cause gradual deterioration of intervertebral discs and facet joints.
Loss of disc hydration and height, reduces shock absorption which increases load on facet joints and causes osteophyte formation.
Secondary nerve root irritation or compression from foraminal narrowing or facet hypertrophy.

The mechanisms of degenerative spinal conditions

- Relatively poor blood supply to the intervertebral discs
- Multiple micro injuries to discs
- Cartilage wear in the joints
- Reduction in the number of chondrocytes
- Vascular ingrowth into the disc
- Marginal osteophytes on vertebral bodies and joints
- Herniation of discs
- Osteopenia and osteoporosis
- Biomechanical overload due to disease at other segments

Clinical clues for degenerative diseases

- Gradual onset of pain and stiffness: pain worsens with activity and improves with rest of mechanical pattern and morning stiffness typically <30 minutes
- Pain and crepitus with movement.
- Localized tenderness.
- Symptoms aggravated by prolonged positions.
- Possible history of previous episodes.
- Reduced segmental mobility.
- Hypomobility on passive accessory movements
- May show evidence of muscle atrophy.
- May have intermittent radicular symptoms (numbness, tingling, shooting pain).
- Absence of constitutional or red flag symptoms.

CERVICAL SPINE DIFFERENTIAL DIAGNOSIS BY ETIOLOGY

D. Inflammatory / Autoimmune





Rheumatoid Arthritis (C1–C2 instability)	Morning stiffness >1 hour, systemic signs, avoid manipulation.
Ankylosing Spondylitis	Pain improves with movement, worse at night, reduced chest expansion.

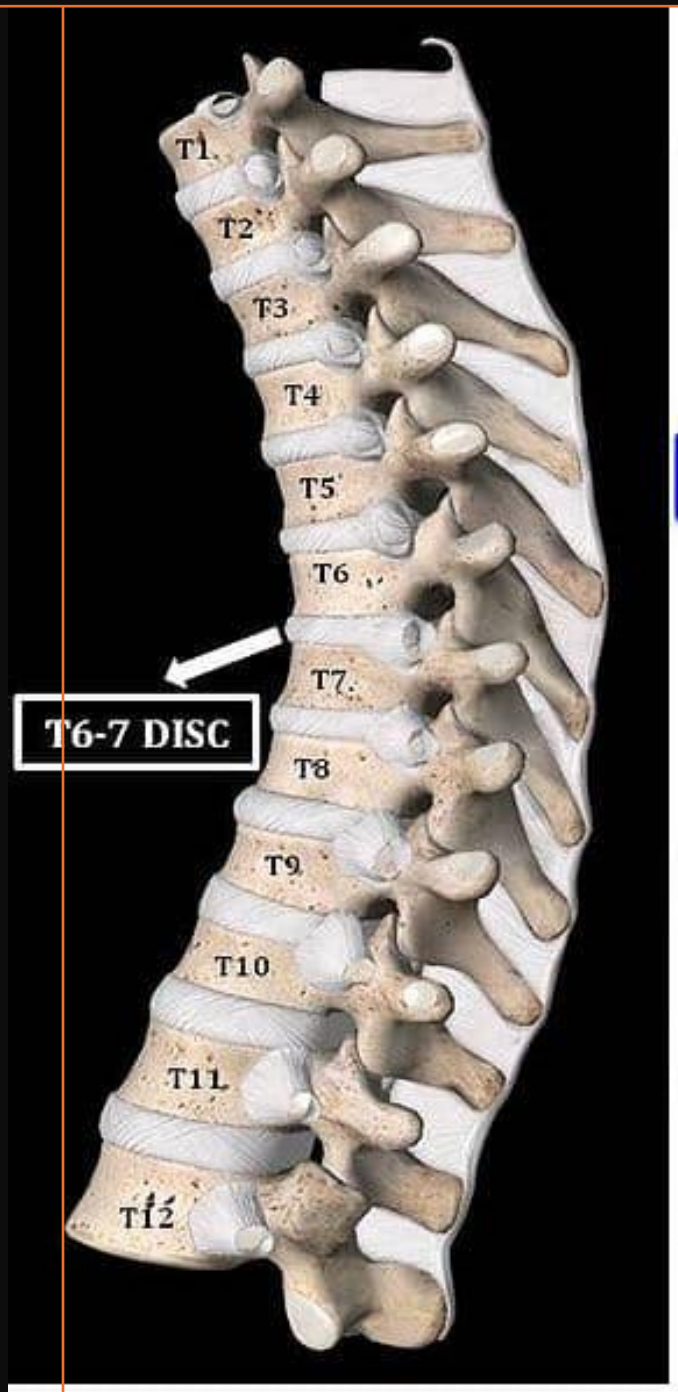
Test your knowledge

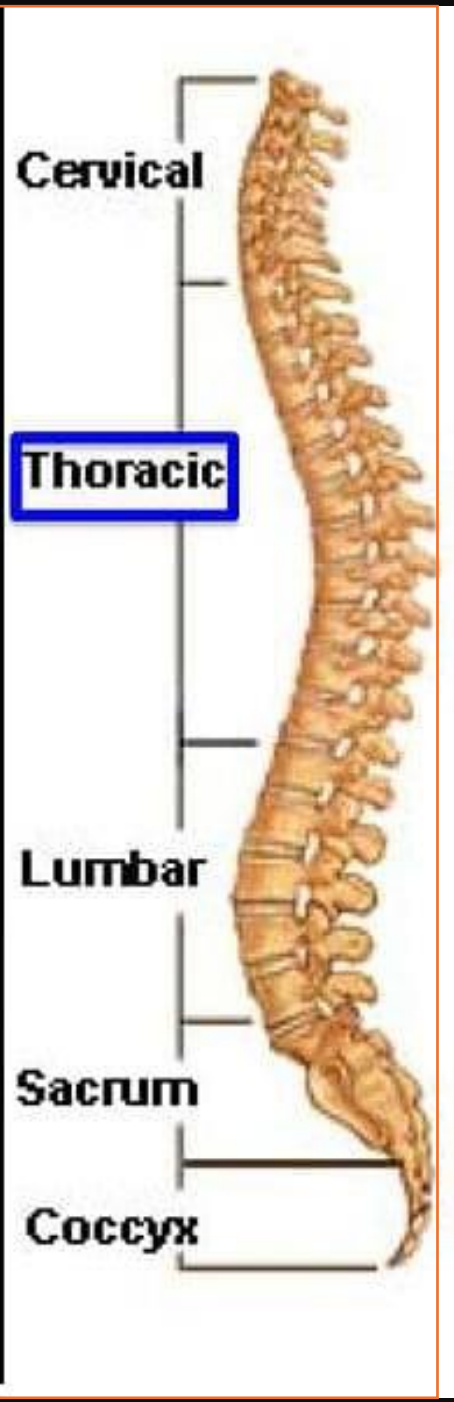
Signs and symptoms	Diagnosis	Confirmation tests
Patient with Unilateral neck pain that is aggravated by Extension + rotation to same side and relieved by: flexion, with No numbness / tingling or neuro deficits	?	Quadrant test (extension + rotation) reproduces LOCAL pain.
Patient with diffuse neck ache, may refer to scapula that is Aggravated by: Flexion, sitting, lifting with Stiffness: morning > evening	?	Pain increases with sustained flexion (prolonged posture)
Pain: Neck → arm → hand (dermatomal) that is Aggravated by: Spurling, Valsalva, compression and Relieved by: Shoulder abduction relief sign (hand on head reduces symptoms)	?	1. Spurling + = nerve root compression 2. Cervical Distraction= Relief pain 3. ULNT1 / Median Nerve Test + = nerve mechanosensitivity * If Spurling + ULNT + Distraction test are all positive → ~90% probability radiculopathy (Wainner cluster).
Pain: Unilateral, starts in neck → wraps around to eye/temple that is aggravated by sustained postures and reduced ROM, especially C1–C2 rotation	?	C1–C2 Flexion-Rotation Test
Bilateral hand or leg paresthesia, Ataxic gait Hyperreflexia in arms and/or legs	?	Hoffmann / Babinski / Clonus +
Bilateral hand or leg paresthesia, No Ataxic gait or Hyperreflexia, But has hyporeflexia in arms	?	

Shoulder Pain Mimics Neck Pain

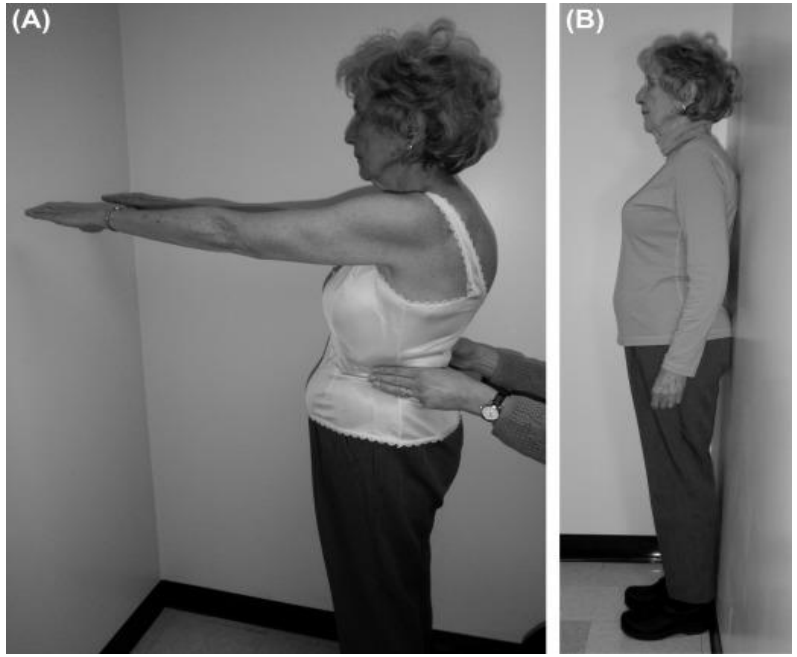
- Rotator Cuff Tear / Tendinopathy: Painful arc in elevation; normal neck ROM
- Subacromial Impingement: Pain with overhead load; no paresthesia
- Frozen Shoulder (Adhesive Capsulitis): Global loss of shoulder ROM, not just painful
- AC Joint Pathology: Pain on cross-body adduction

Thoracic spine

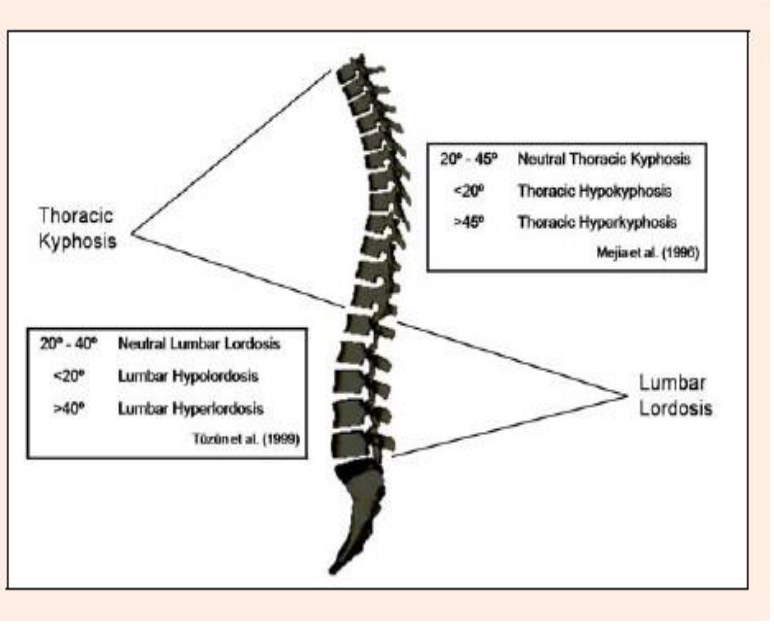




Mechanical Musculoskeletal Thoracic Pain: C) Thoracic Kyphosis Syndrome



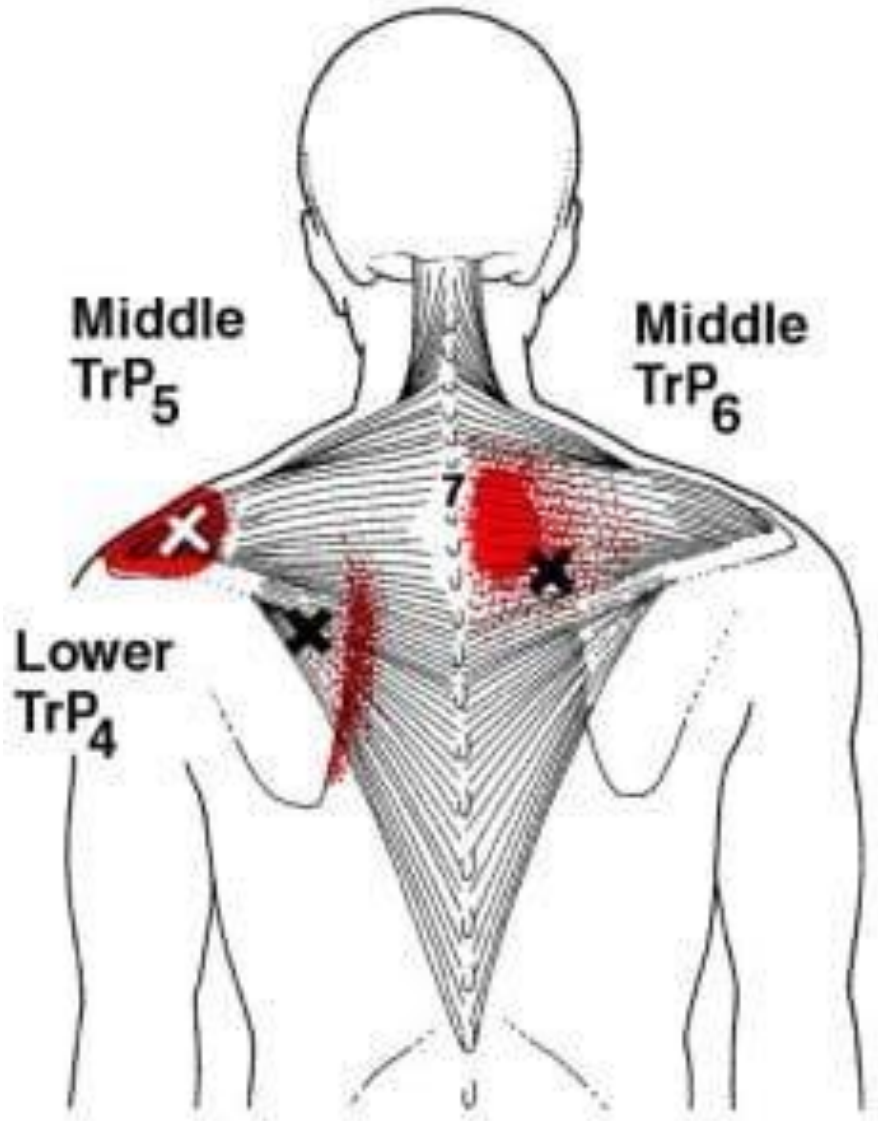
- involves an exaggerated forward (anterior) curvature of the thoracic spine, typically greater than 50 degrees. Normal is between 20-45 degrees.
- Common in desk work, students, or post-op patients.
- Dull, aching mid-back fatigue and worse with prolonged sitting
- Improves with thoracic extension exercises
- No referral below ribs.
- Muscle Imbalances: The sustained poor posture leads to specific muscle imbalances:
    - Lengthened/Weakened Muscles: Erector spinae (upper thoracic/lumbar), abdominal muscles, and scapular stabilizers become overstretched and less effective.
    - Shortened/Tight Muscles: Chest (anterior) muscles become shortened, making it difficult to maintain proper shoulder position and take deep breaths.



Mechanical Musculoskeletal Thoracic Pain: D)Myofascial Pain Syndrome

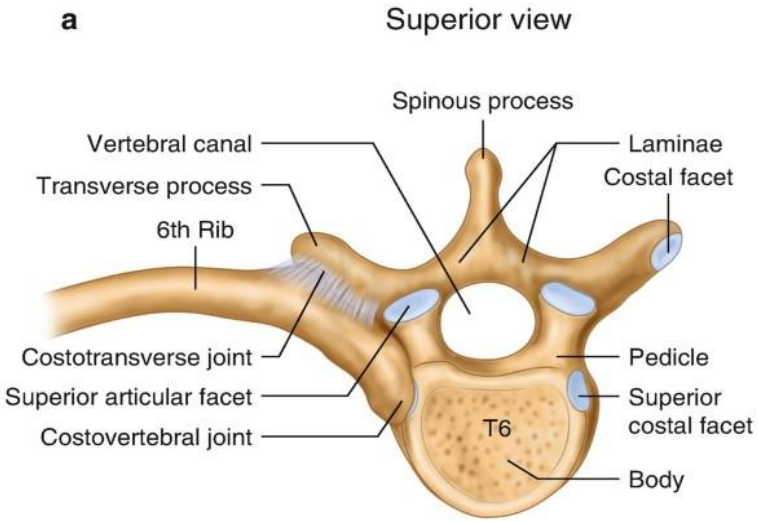
Involving: Trapezius, Rhomboids, Paraspinals, Levator scapulae and Serratus posterior superior

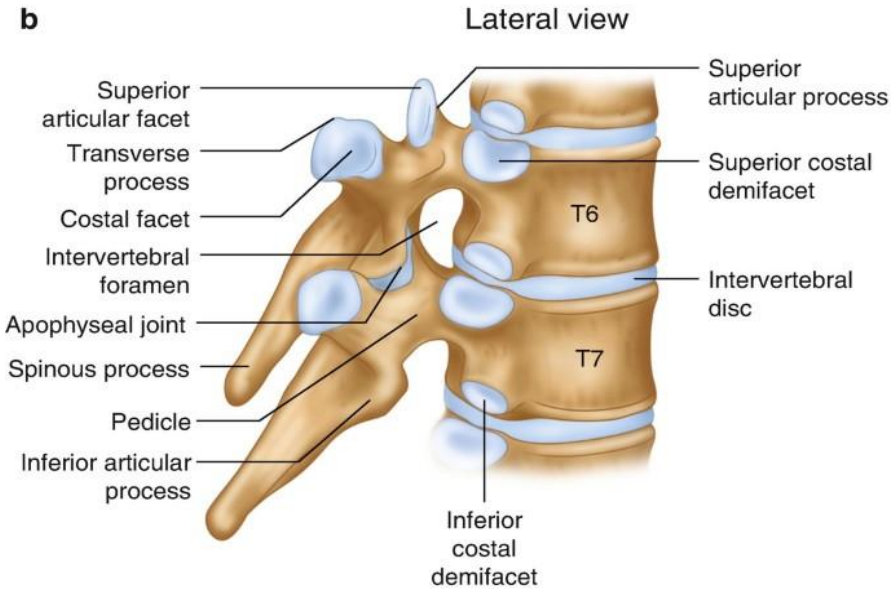
- Trigger points may refer pain to scapula or chest wall.



DD of Thoracic spine pain: Functional Anatomy

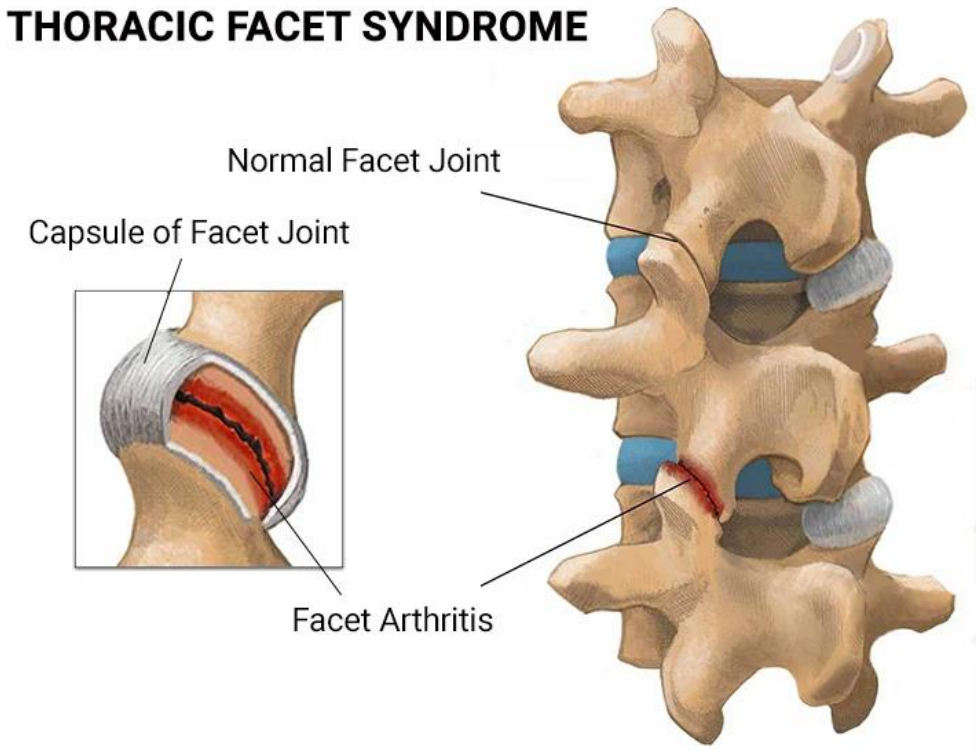
Structure	Clinical Relevance
Facet (Zygapophyseal) Joints	Source of localized thoracic pain; aggravated by extension + rotation.
Costovertebral & Costotransverse Joints	Pain increases with breathing, coughing, or rib compression.
Thoracic Discs	Rarely herniate (due to rib stability), but when they do → band-like chest or abdominal pain.
Intercostal Nerves	Can cause sharp, dermatomal chest-wall pain (T4–T10 shingles or nerve entrapment).





1- **Mechanical Musculoskeletal Thoracic Pain: A) Thoracic Facet (Zygapophyseal) Joint Dysfunction**

- Pain in Thoracic facets can be caused by repetitive rotation/extension, static posture. Pain is aggravated with extension + rotation and local palpation
- Localized unilateral thoracic pain, No neurological symptoms



Mechanical Musculoskeletal Thoracic Pain: B) Costotransverse / Costovertebral Joint Dysfunction

This issue is often misdiagnosed as muscle strain which is caused by rib irritation or restricted rib glide.

Clinical signs:

Pain worse with deep breath, cough, sneeze and may travel along rib toward sternum

Can be tested by pain reproduced by rib spring test



2) NEURAL & DISCOGENIC CONDITIONS

Thoracic Radiculopathy (Disc or Nerve Root)

- Rare but important, comes with sharp / burning pain that wraps around the chest like a band (T4–T10).
- Often unilateral
- Worsens with flexion or cough
- Minor dermatomal sensory changes possible
- Must rule out *cardiac or visceral pain first*.

Intercostal Neuralgia

- neuropathic pain in the intercostal nerves, which run along the ribs, chest, and abdomen.
- It is characterized by sharp, aching, burning, or stabbing pain along rib space, sometimes with numbness or tingling, often worse with movement like breathing, coughing, or sneezing.
- Causes include nerve damage from trauma, surgery, or viral infections like shingles.
- Aggravating factors: Rotation, Rib compression and Inspiration

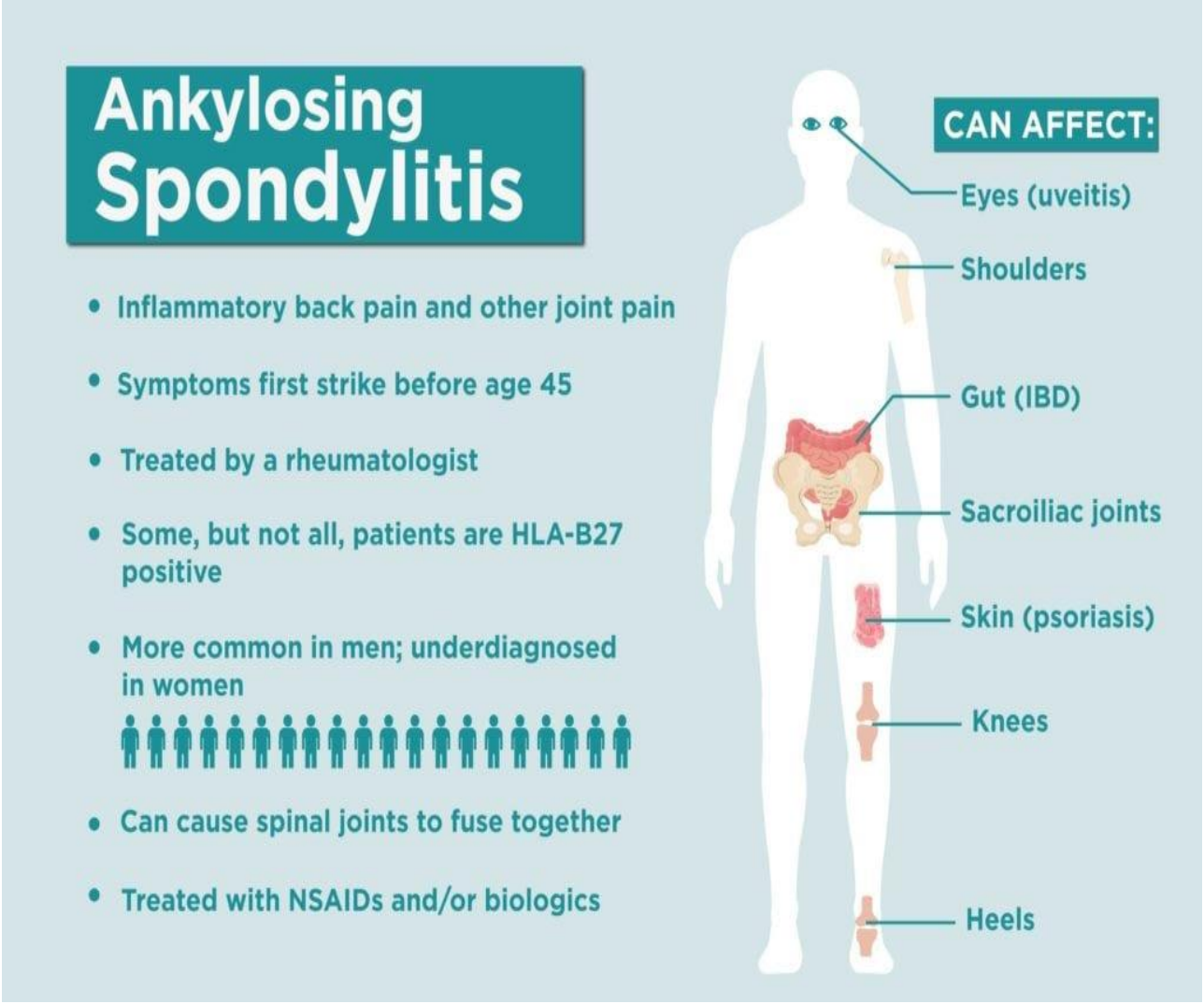
3) Structural: Scheuermann’s Disease (Adolescents)

- It is a spinal disorder that develops in adolescents, causing an excessive forward curvature (hyperkyphosis) of the upper back, resulting in a hunched appearance.
- It is characterized by wedge-shaped vertebrae, which cause the spine to bend and can lead to back pain and poor posture.
- Caused by either genetic or idiopathic abnormal vertebral growth where the front of the vertebrae doesn't grow as fast as the back, causing them to become wedge-shaped instead of rectangular.
- Visible thoracic rounding
- Hamstring tightness
- Fatigue-based back pain



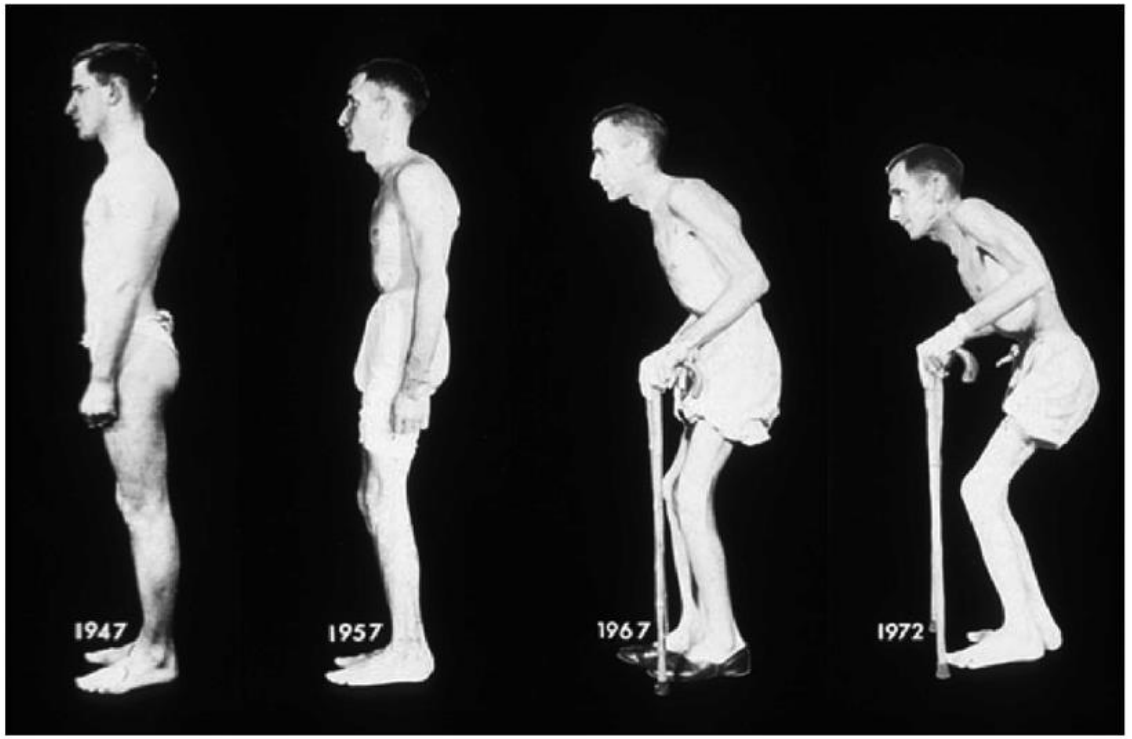
Inflammatory Conditions: Ankylosing Spondylitis (AS)

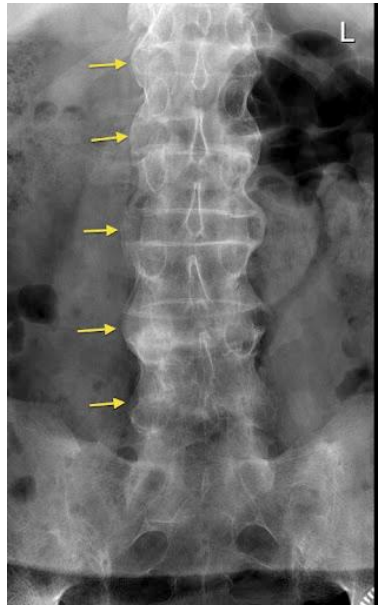
- This inflammatory disorder is the most frequent seronegative inflammatory spinal disease in adults and the prototype for other members of the spondyloarthropathies.
- The prevalence of this disorder is estimated at 0.1–0.2% in the general population. Ankylosing spondylitis primarily affects young males, with a male-to-female ratio between 4:1 and 10:1. The mean age of presentation is 26 years.
- The HLA-B27 gene (human leukocyte antigen B27) is present in 95% of patients, but only a small percentage of all individuals positive for HLA-B27 actually develop the disorder.



Ankylosing Spondylitis (AS)

- AS typically begins in the sacroiliac joints and then affects the thoracolumbar and lumbosacral regions of the spine. Eventually, the spine becomes completely rigid, with loss of its normal curvatures and movement.
- Untreated, it renders the spine at risk of transverse fractures following relatively minor injury.
- In the late stage, squaring of the vertebral bodies is a characteristic feature of AS and is caused by a combination of corner erosions and periosteal new bone formation along the anterior aspect of the vertebral body.
- Further inflammation is associated with formation of syndesmophytes: ossifications of the outer fibres of the annulus fibrosus that form bony bridges between the inflamed corners of adjacent vertebrae.
- Ossification also occurs within the fibres of the adjacent paravertebral connective tissue and of the posterior interspinous ligaments.
- Complete fusion of the vertebral bodies by syndesmophytes and related ossification produces the “bamboo spine” where the bone mimics a bamboo cane with segments of thickening.
- Fractures in AS usually occur at the thoracolumbar and cervicothoracic junctions where the biomechanical load is greatest. Spinal fractures are characteristically transverse (i.e., horizontal) rather than vertical or oblique and are called “chalk stick” or “banana” fractures. This is because these items will break transversely if bent.





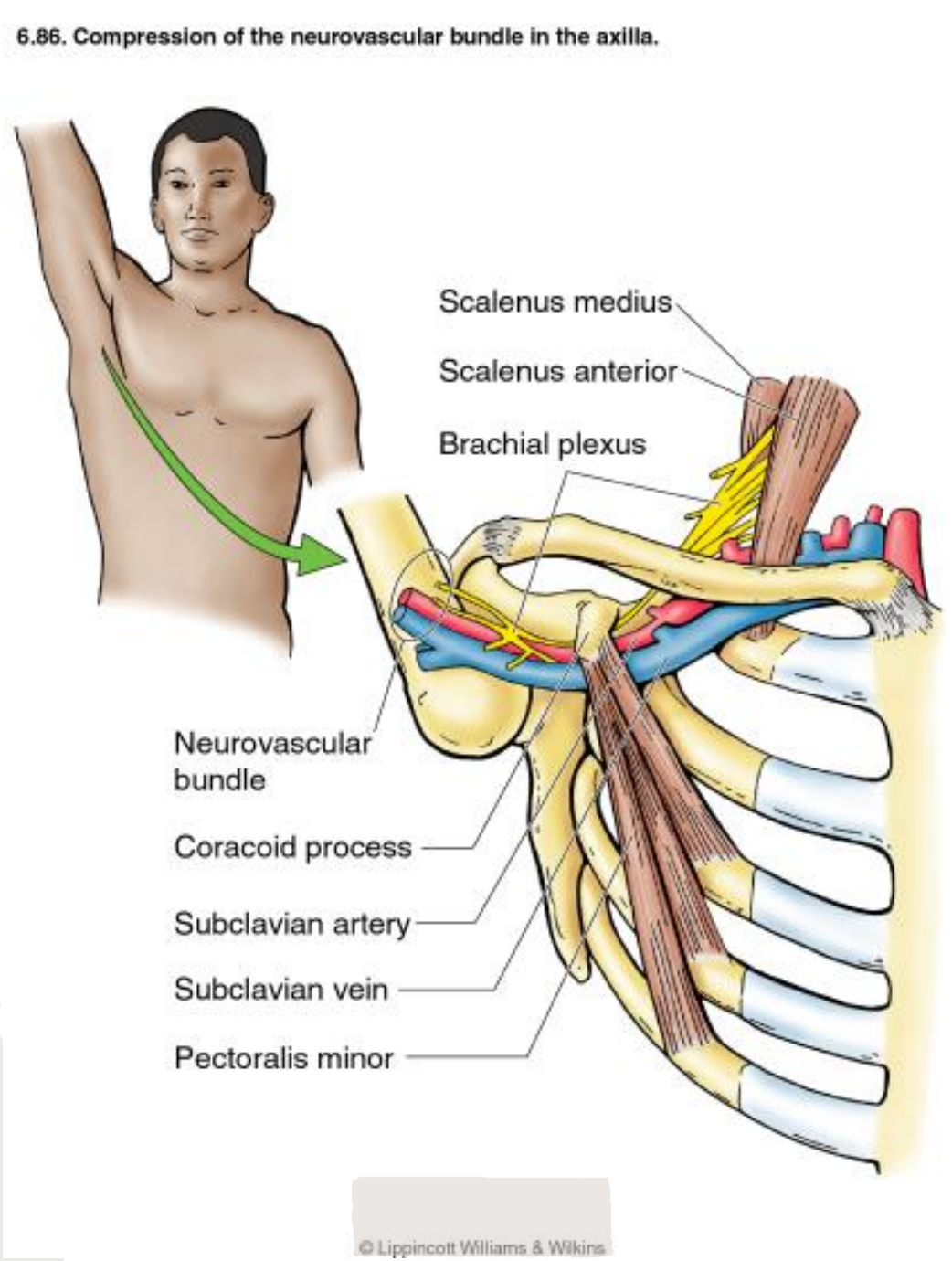
5) Viseral Referral pain

- This happens because the brain misinterprets the pain signal, as the internal organ and the perceived pain location share the same spinal cord segments for nerve signals. This can cause pain from the abdomen, for instance, to be felt in the chest or back.

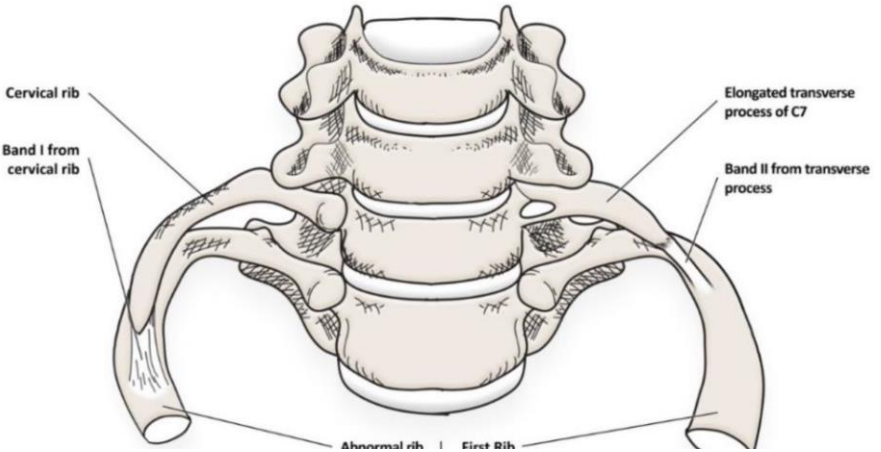
- Heart (Angina/MI), Left thoracic or chest pain ± arm/jaw and worse with exertion, not palpation
- Gallbladder, Right scapular or mid-thoracic pain Worse after fatty meals
- Stomach / Duodenum Central thoracic pain Better/worse with eating
- Pancreas Midline thoracic → epigastric Worse lying supine
- Kidney Thoracolumbar junction pain on flank percussion

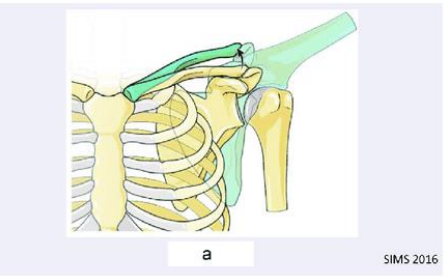
Vascular : Thoracic Outlet Syndrome (TOS)

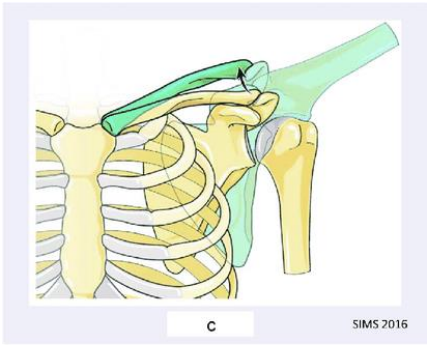
- TOS refers to the compression of the neurovascular structures, primarily the brachial plexus, subclavian artery, and the subclavian vein, as they exit through the thoracic outlet also known as the cervicothoracobrachial region.
- Anatomically, the thoracic outlet is bordered anteriorly by the anterior scalene muscle, posteriorly by the middle scalene , and inferiorly by the first rib .
- This area serves as a passageway for nerves and blood vessels travelling from the neck to the arm.
- Clinically it is typically divided into two main categories:
  - Vascular TOS, which may involve compression of the subclavian artery or vein and generally presents with fewer diagnostic problems.
  - Neurogenic TOS which involves compression of the brachial plexus and accounts for more than 90% of all TOS cases.

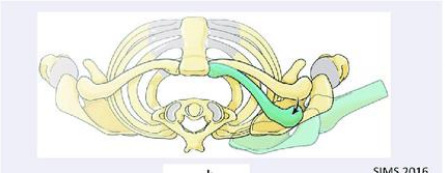












elevation/lowering (a), protraction-retraction (b) and anterior/posterior axial rotation (c).

Causes of TOS

Several congenital or acquired anatomical abnormalities can further compromise the thoracic outlet, including:

- The presence of a cervical rib (Long C7 transverse process)
- Fibrous bands or muscular anomalies (e.g. a broad scalene attachment or hypertrophied scalenes)
- Clavicular hypomobility
- Scoliosis or elevated scapula
- Functionally acquired anatomical changes.

Types of TOS



Arterial TOS (ATOS)

- Young adult with vigorous arm activity
- Pain in the hand, Claudication, Pallor, Cold intolerance, and Paresthesia

Venous TOS (VTOS)

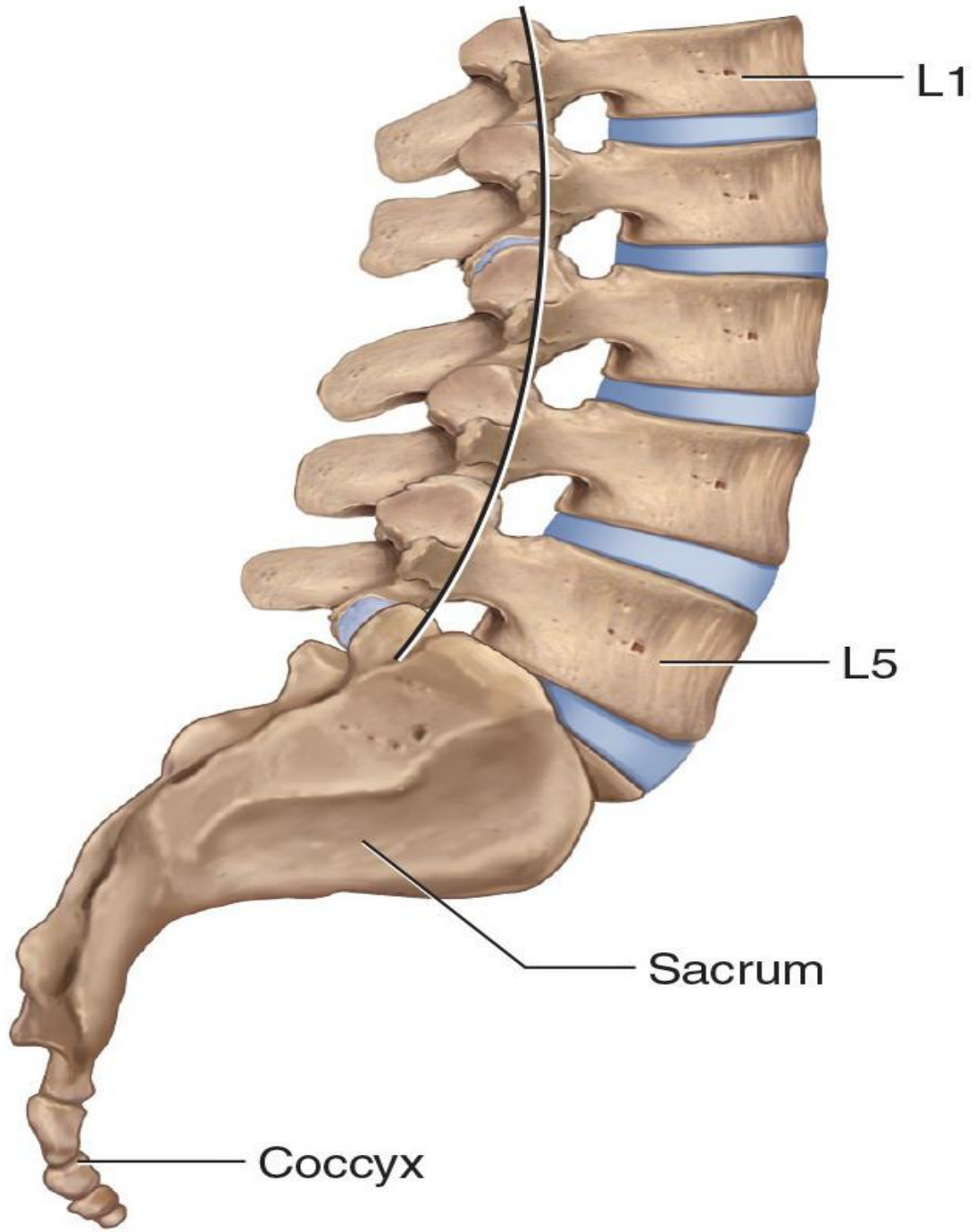
- Younger men involved in strenuous upper limb use
- Cyanosis, Feeling of heaviness, Paresthesia in fingers/hand (due to oedema), Arm swelling (oedema)



True Neurogenic TOS

- History of trauma (e.g. whiplash)
- Pain, numbness, paresthesia, and/or weakness, Day/night symptoms, Loss of fine motor skills, Objective weakness

Lumbar spine



Mechanical LBP

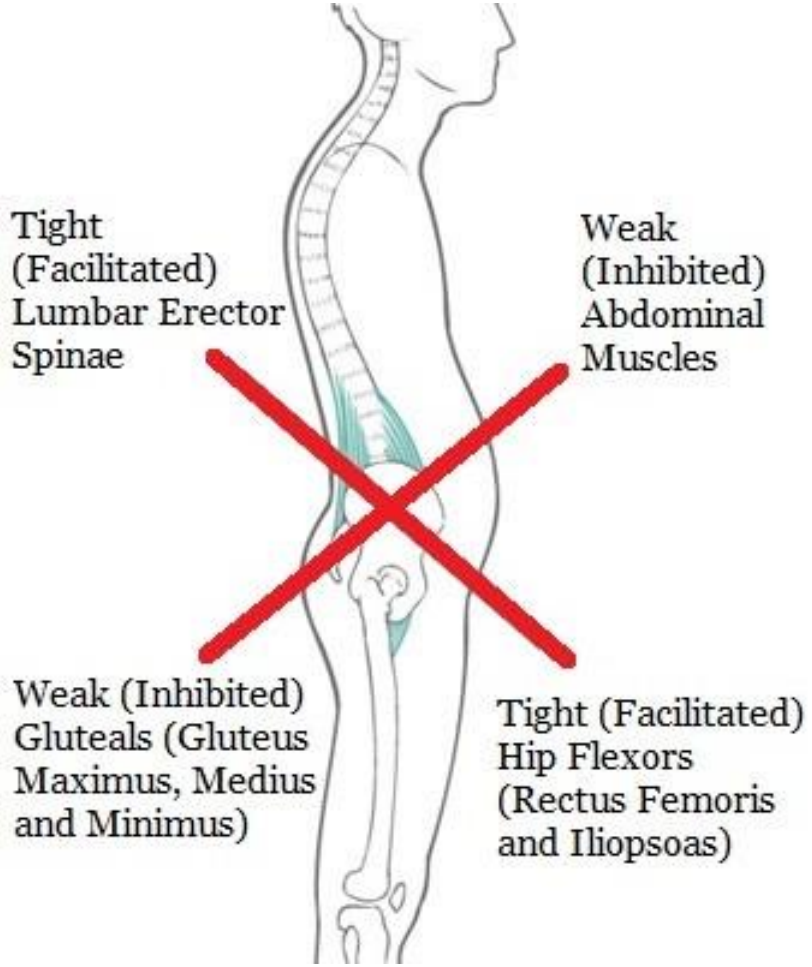
- Mechanical back pain is pain which arises from a structure within the spine, including the vertebral bodies, intervertebral discs, zygapophysial joints, sacroiliac (SI) joints, spinal ligaments, paraspinal muscles, dura, spinal cord, and nerves

Possible sources which may present with mechanical back pain[9]

- Degenerative disc diseases
- Facet joint derangement
- Osteoarthritis
- Muscle imbalance
- Spinal stenosis
- Herniated disc
- Spondylolisthesis
- Fracture
- Severe kyphosis
- Severe scoliosis
- Transitional vertebrae

Possible sources which may present with inflammatory back pain[7,9]

- Inflammatory arthritis
  - Axial Spondyloarthritis including Ankylosing Spondylitis (AS)
  - Psoriatic arthritis (PsA)
  - Inflammatory bowel disease (IBD)
  - Psoriatic spondylitis
  - Reiter's syndrome



Lower Crossed Syndrome

- Lower Crossed Syndrome is characterised by a specific pattern wherein certain muscle groups demonstrate increased activity and perceived "tightness" (hip flexors, particularly iliopsoas and tensor fasciae latae; and lumbar extensors including erector spinae and quadratus lumborum), while antagonistic muscles show reduced activity and apparent "weakness" (deep abdominal muscles including transversus abdominis, and gluteal muscles, particularly gluteus maximus and medius). The hamstrings are frequently observed to have increased tone in this pattern as well.

Posture-Pain Relationship

- A 2024 systematic review and meta-analysis examining postural asymmetry in low back pain found that whilst pelvic tilt was statistically higher in participants with low back pain compared to controls, the clinical significance of these differences remains unclear.
- The authors noted substantial heterogeneity in measurement methods and concluded that lumbopelvic mechanisms may be altered in people with low back pain, but that no firm conclusions could be drawn regarding causation.
- Importantly, postural variations exist across large populations of pain-free individuals, suggesting that specific postural alignments cannot be considered inherently pathological.

- Research on lumbar lordosis presents conflicting findings. A 2017 meta-analysis found that patients with low back pain tended to have decreased lumbar lordosis compared to healthy controls, particularly in cases involving disc herniation or degeneration—a finding that contradicts the increased lordosis typically associated with Lower Crossed Syndrome.
- However, the studies included were heterogeneous, and factors such as age, pain severity, and measurement methodology significantly influenced results. This highlights the individual variability in spino-pelvic alignment and the limitations of assuming a single "ideal" posture.

Muscle Activation and Motor Control

- The concept of specific muscles being "inhibited" or "facilitated" in predictable patterns oversimplifies the complexity of motor control. The central nervous system selects muscle activation strategies based on multiple factors including task demands, previous experience, perceived threat, fatigue, and pain.
- What may appear as "weakness" in clinical testing could represent altered motor planning, pain-related guarding, or task-specific adaptation rather than true muscle inhibition.
- Similarly, perceived "tightness" may reflect neural tension, increased muscle tone secondary to guarding, or altered sensory perception rather than shortened muscle tissue.
- Recent research examining motor control in individuals identified as having Lower Crossed Syndrome suggests alterations in gait patterns and postural control, but the direction of causality remains unclear.
- Do muscle imbalances cause pain and dysfunction, or do pain and altered movement patterns lead to observable changes in muscle activation? Current evidence suggests the latter is equally, if not more, plausible.

Lumbar Spine: Spondylosis (Degenerative Spinal Osteoarthritis)

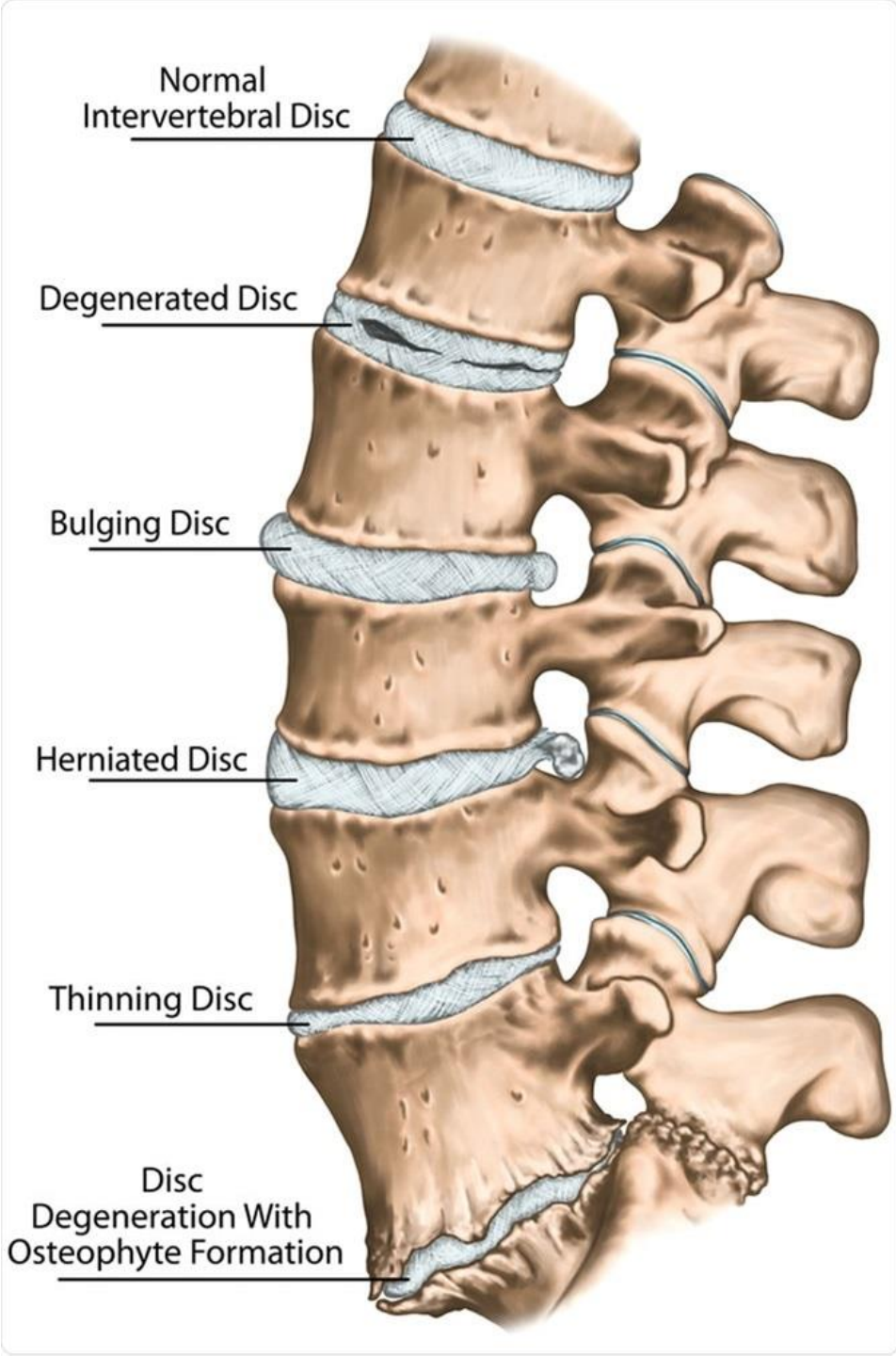
- Spondylosis is degenerative change of the lumbar spine that reflects age-related wear, but symptoms arise when degeneration alters load distribution.
- Degenerative changes can affect the intervertebral discs and facets.

Distinguishing Characteristics

- Broad stiffness, not sharp or radiating unless nerve root is involved
- Pain is often mechanical and predictable
- No inflammatory pain pattern (doesn’t improve with exercise)
- May be affected by temperature.

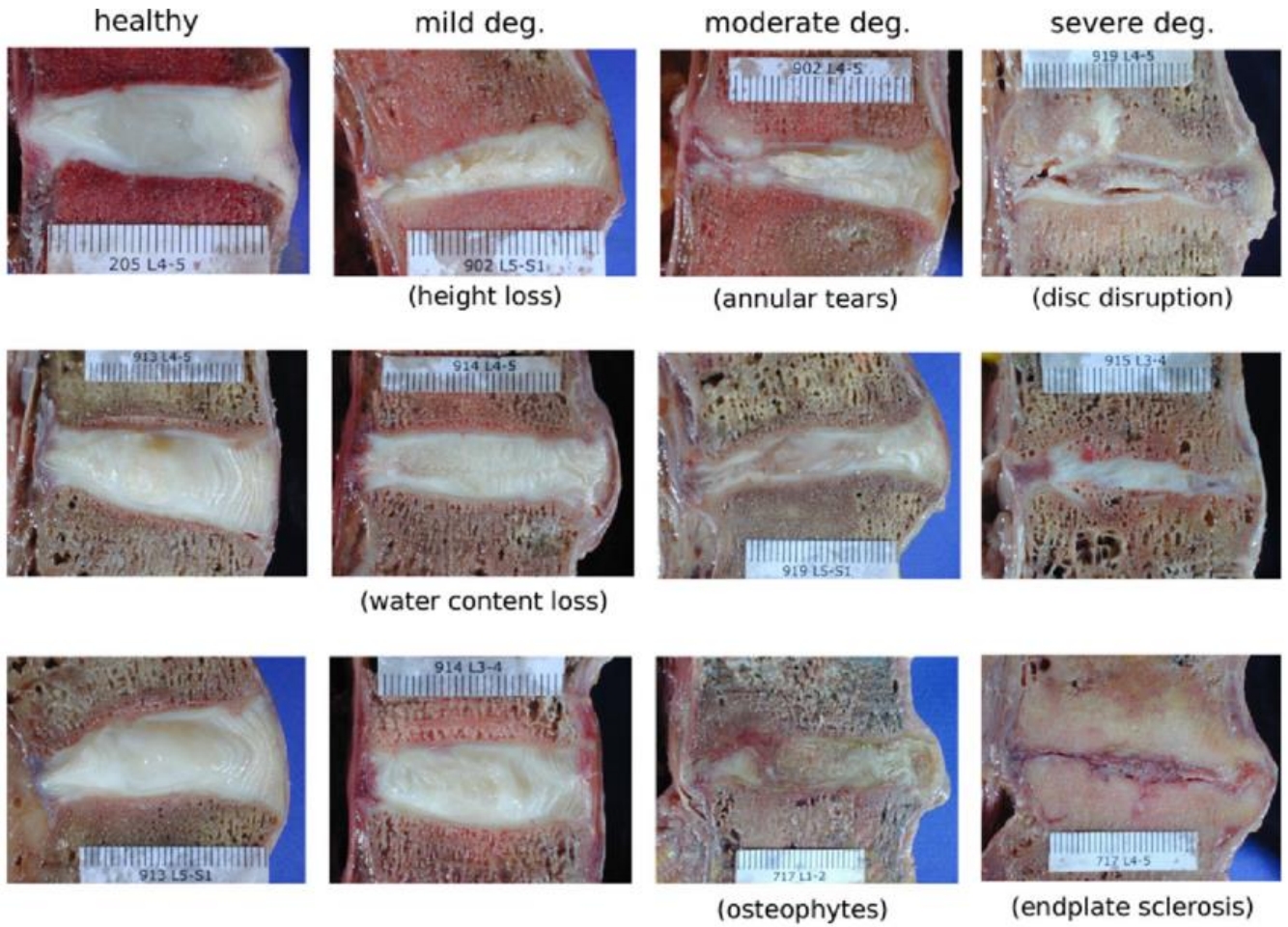
Degenerative Intervertebral Disc

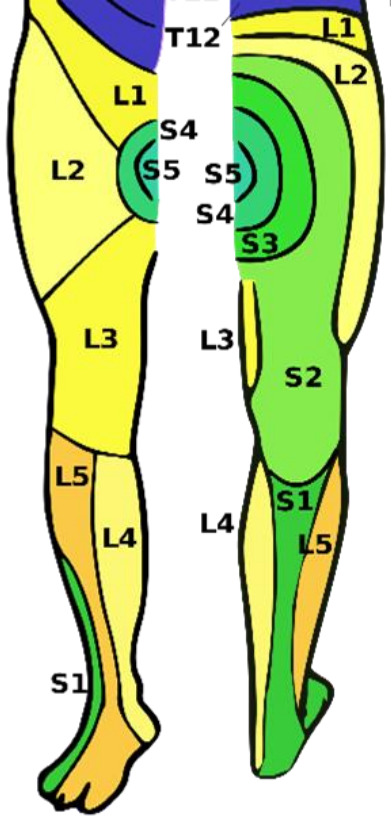
- The lumbar disc is the primary structure that absorbs compressive load in the spine. It becomes painful when it is stressed by flexion-based loading, such as bending forward, prolonged sitting, or lifting with a flexed spine.
- When irritated, the disc produces deep, central low back pain, and may refer pain into the buttock or posterior thigh, though it typically does not radiate below the knee unless a nerve root becomes involved.

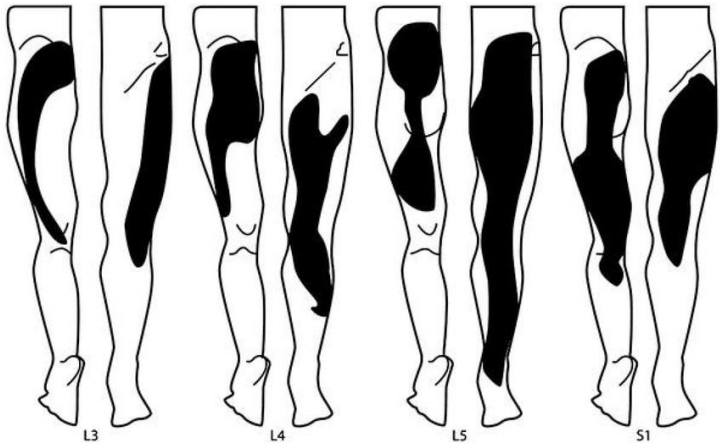


Degenerative Intervertebral Disc

- Pain is deep central or paracentral low back pain, midline tenderness is common, but not always diagnostic can be aggravated by flexion-based postures (sitting, bending, slumping),morning movement as the disc hydration at night increases pressure, coughing/sneezing as it increases the intra-discal pressure.
- Pain is eased by standing and gentle extension, and short bouts of walking
- Reduced lumbar lordosis, antalgic posture (often lateral shift in acute herniation)







Lumbar Conditions: Nerve Root / Radiculopathy

Nerve Pain: Sharp, burning, or electric leg pain > back pain

- **Myotomes:**

L4: Knee extension weakness

L5: Big toe extension weakness

S1: Plantarflexion weakness

- **Dermatomes:**

Clear dermatomal pattern

Dermatomal numbness/tingling

- Reflex :Patellar ↓ (L4)Achilles ↓ (S1)Root / Radiculopathy

Special Tests for Lumbar Radiculopathy

Straight Leg Raise (SLR): Reproduction of leg pain below knee L4–S1 nerve root tension / disc herniation.

Crossed SLR: Opposite leg movement reproduces symptoms, Strong indicator of disc herniation.

Slump Test: Reproduction of leg symptoms with neural sensitization, Neural mechanosensitivity / radiculopathy.

Repeated Flexion in Standing: Symptoms worsen or peripheralize, Flexion-sensitive disc pathology.

Femoral Nerve Stretch: (L2–L4)Upper lumbar radiculopathy

Musculoskeletal Lumbar Conditions: 2) Facet (Zygapophyseal) Joint Pain

- Pain is unilateral paraspinal, sometimes into posterior hip or upper lateral thigh can be aggravated by extension, rotation, prolonged standing walking downhill or descending stairs
- Eased by sitting, flexion, leaning forward
- Neurological findings: normal (no sensory/motor/reflex changes)

Special tests

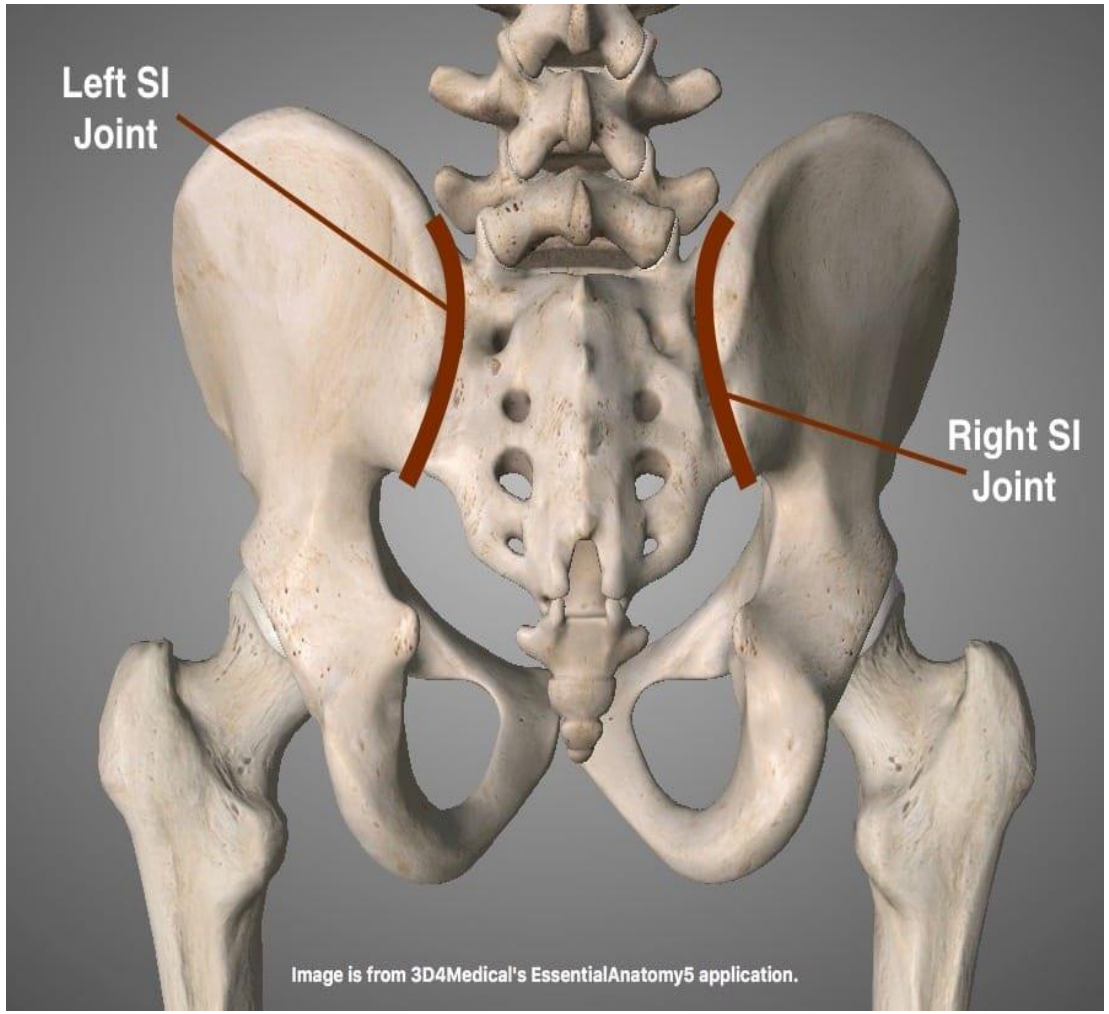
- Extension-rotation test (kemp's), local pain at facet joint (not leg pain), facet irritation / facet arthropathy
- PA glide over facet, reproduction of familiar pain segmental facet involvement
- Repeated extension movements and pain worsens without referral change, extension sensitivity: facet dominant
- Facet pain never radiates below the knee.

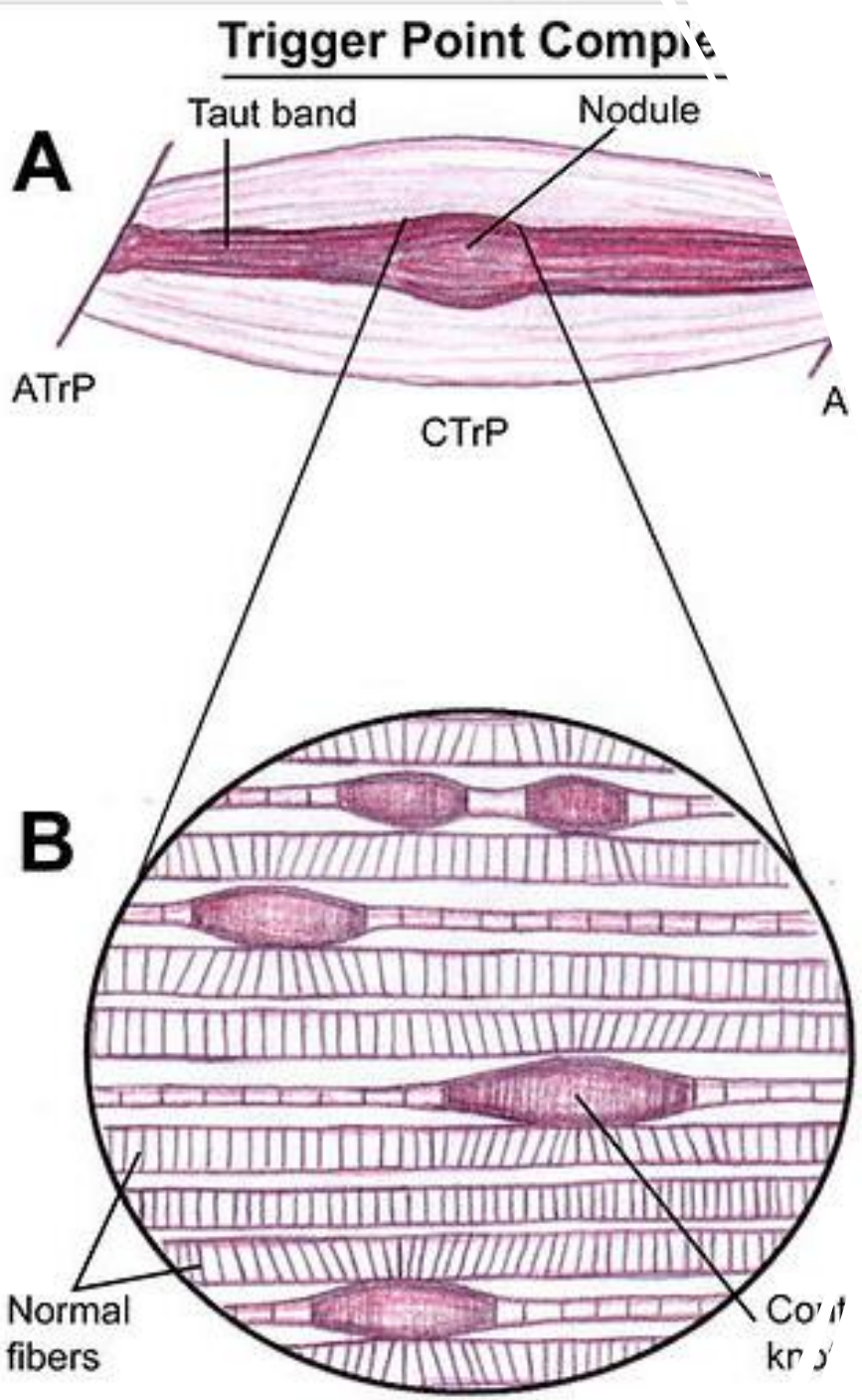
.

Musculoskeletal Lumbar Conditions: 3- Sacroiliac Joint (SIJ) Pain

- Pain felt just medial to PSIS, aggravated by walking, stairs, transitional movements (sit to stand), standing on one leg
- Referral pattern to posterior thigh or groin (common, often misinterpreted as hip or lumbar)
- SIJ special tests: use a cluster diagnosis is supported when ≥3 of these are positive:

1. Sij pain does not follow a dermatome
2. Thigh thrust
3. Gaenslen's
4. Faber
5. Compression / distraction
6. Stork / gillet test



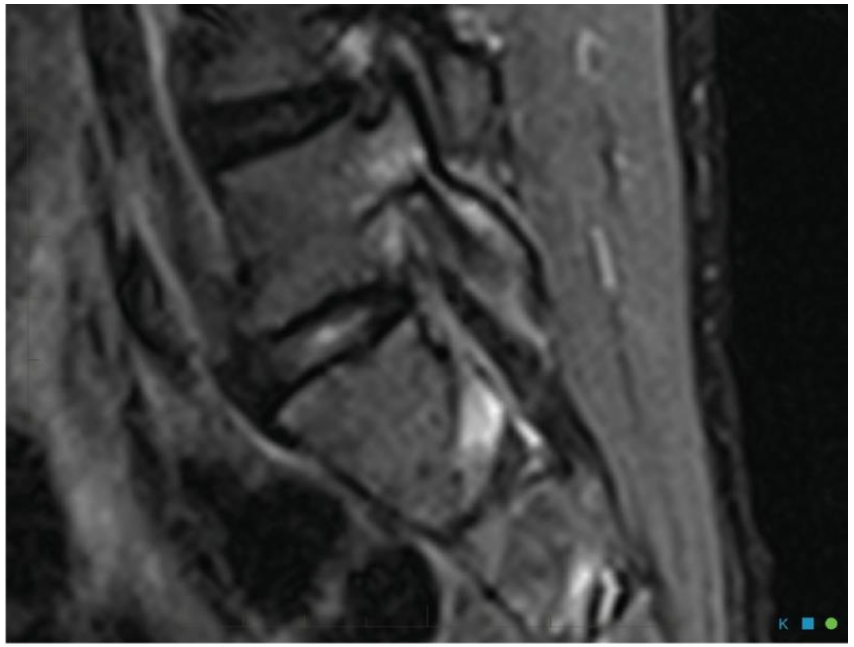


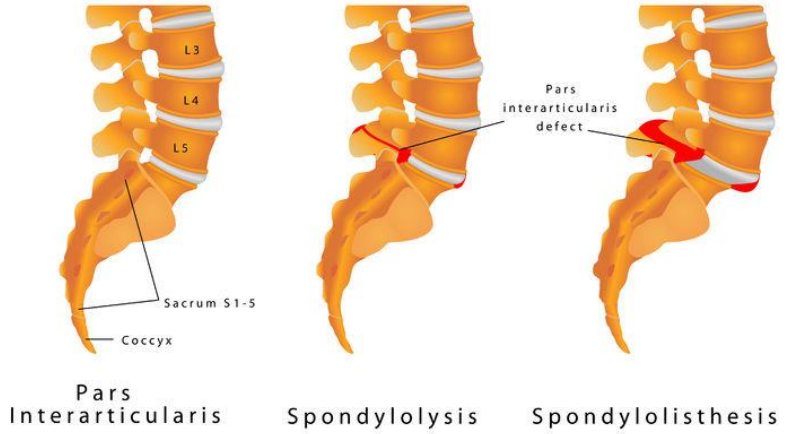
Lumbar Conditions: Myofascial Pain Syndrome

- Aggravated by prolonged postures and rapid load change (lifting poorly) and palpation of a thickened, guarded, or hypertonic paraspinals. Pail at end of full range of motion
- Normal neurological exam
- Essential diagnostic criteria:
  - Hypersensitive spot
  - Taut band
  - Recognised referred pain pattern
- Supporting diagnostic criteria:
  - Local twitch response (LTR)
  - Reduced range of motion
  - Reproduction of symptoms

Pars Interarticularis

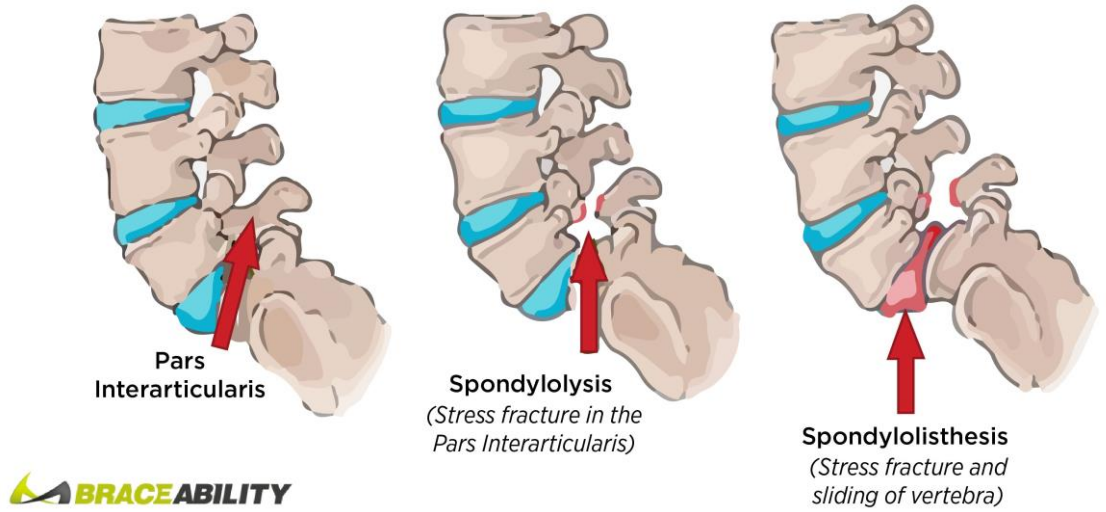
- Pars interarticularis stress fractures are common in young people. Typically, they occur in adolescents who are undertaking sport or physical activity that places repeated stress on bone which is not yet mature.
- MRI demonstrates bone oedema for some weeks before a stress fracture occurs.
- This is an important diagnosis to make because even a few days of additional activity may irretrievably fracture the bone. This can lead to a lifetime of back pain and disability.
- Strict exercise limitation for 6 weeks followed by a careful rehabilitation can lead to complete and permanent healing of the stress response.





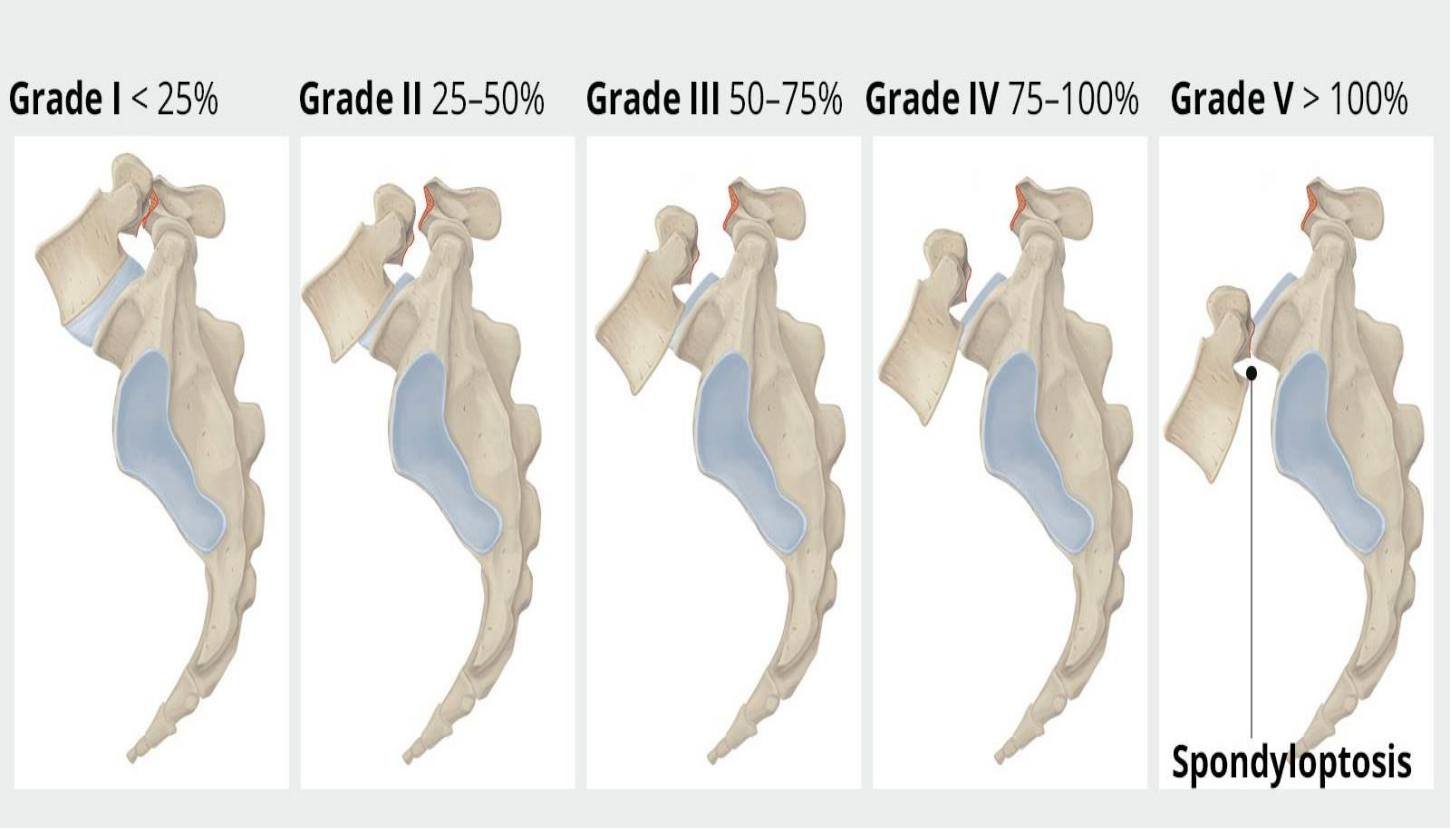
Lumbar Spondylolysis

- Spondylolysis and spondylolisthesis has a high prevalence in sporting and athletic populations.
- A fatigue stress fracture of the pars interarticularis, often at L5.
  Common in adolescents, gymnasts, dancers, cricketers, footballers high repetitive hyperextension loading.
- Pathophysiology: repeated lumbar extension + rotation leads to microfractures of the pars.
- Pain is localized low back pain (often unilateral), aggravated by extension activities (running, kicking, jumping), and eased by flexion, and rest
- Neurological findings usually absent unless progression → slip



Spondylolisthesis

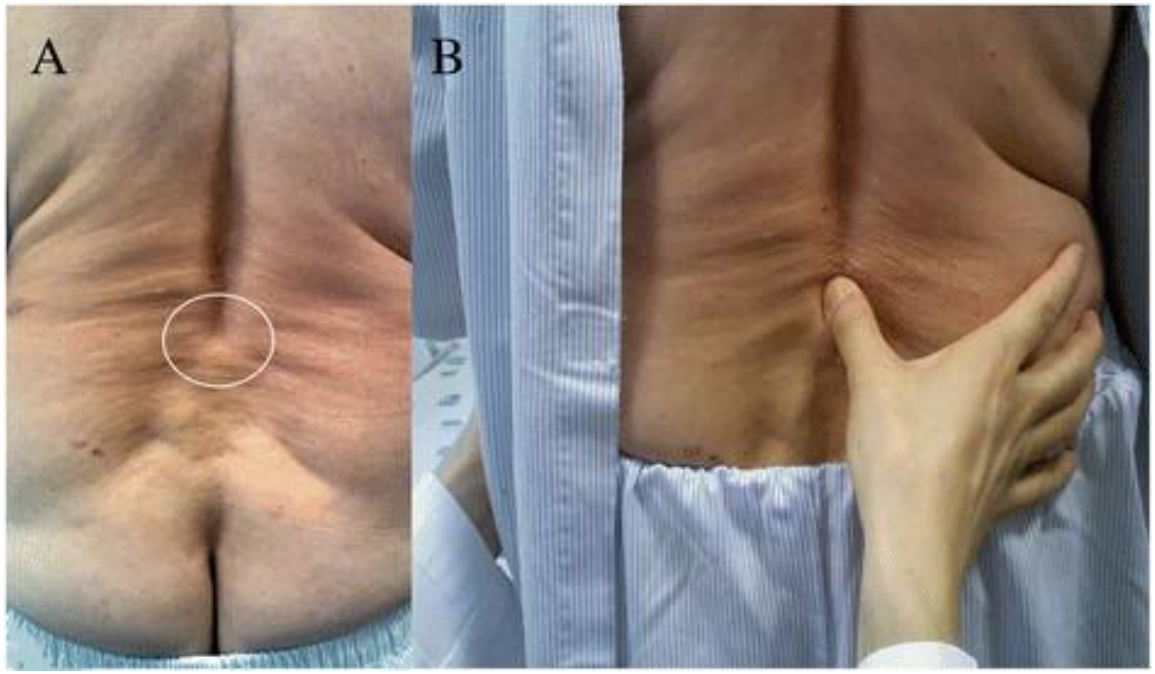
- Spondylolisthesis occurs when a vertebra slips forward, most commonly:
- L5 on S1 (isthmic slip from spondylolysis)
- L4 on L5 (degenerative slip in older adults)

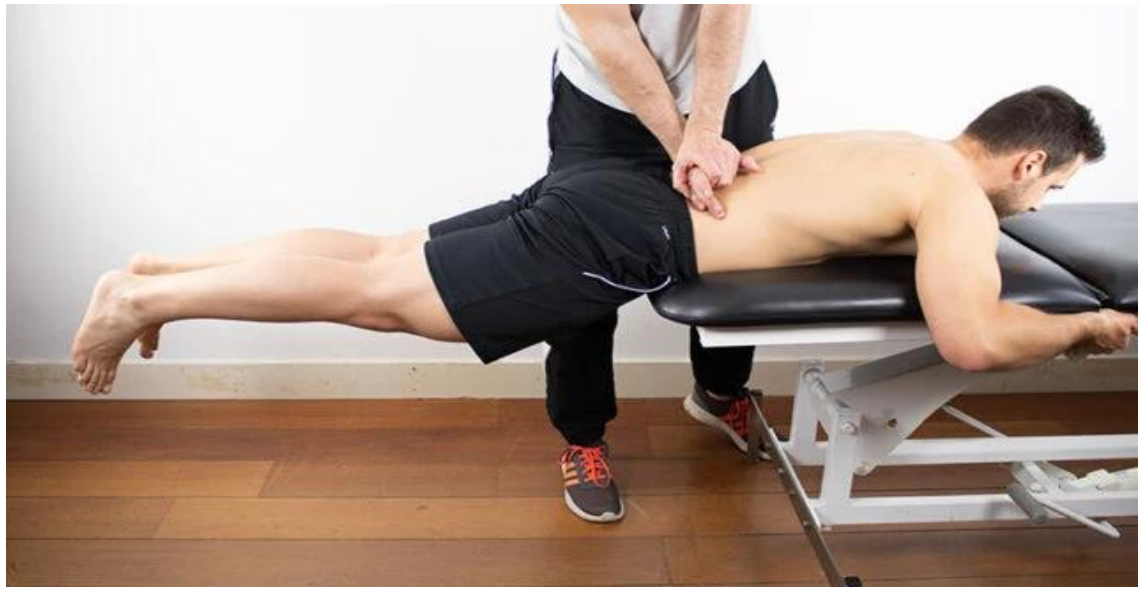


Grade	% Slip
I	0–25%
II	26–50%
III	51–75%
IV	>75%

Spondylolisthesis

- Pain pattern low back pain ± buttock pain
- Aggravated by extension prolonged standing, walking and eased by flexion (leaning forward)
- Referral possible radiating leg symptoms if nerve root compressed
- Palpation step-off deformity at spinous processes (especially grade II+)
- Step Deformity Palpation: Palpable “drop” at spinous process Suggests vertebral slip.
- Prone Instability Test: Pain reduces when paraspinals engage (legs lifted) Indicates segmental instability





Clinical tests for spondylosis

- The one-legged hyperextension test demonstrated low to moderate sensitivity and low specificity to diagnose spondylolysis.
- Lumbar spinous process palpation had high specificity and was the optimal diagnostic test for lumbar spondylolisthesis.
- A variety of diagnostic imaging methods are used to identify the presence of spondylolysis, including plain-film imaging, computed tomography (CT), magnetic resonance imaging (MRI), single photon emission computed tomography (SPECT) and bone scintigraphy.



Pseudo-Sciatica

- Pain that mimics sciatica (pain down the leg), but does NOT come from lumbar nerve root compression.
- In pseudo-sciatica, the sciatic nerve is irritated outside the spine, usually by muscular or biomechanical causes, not the disc or foraminal stenosis.

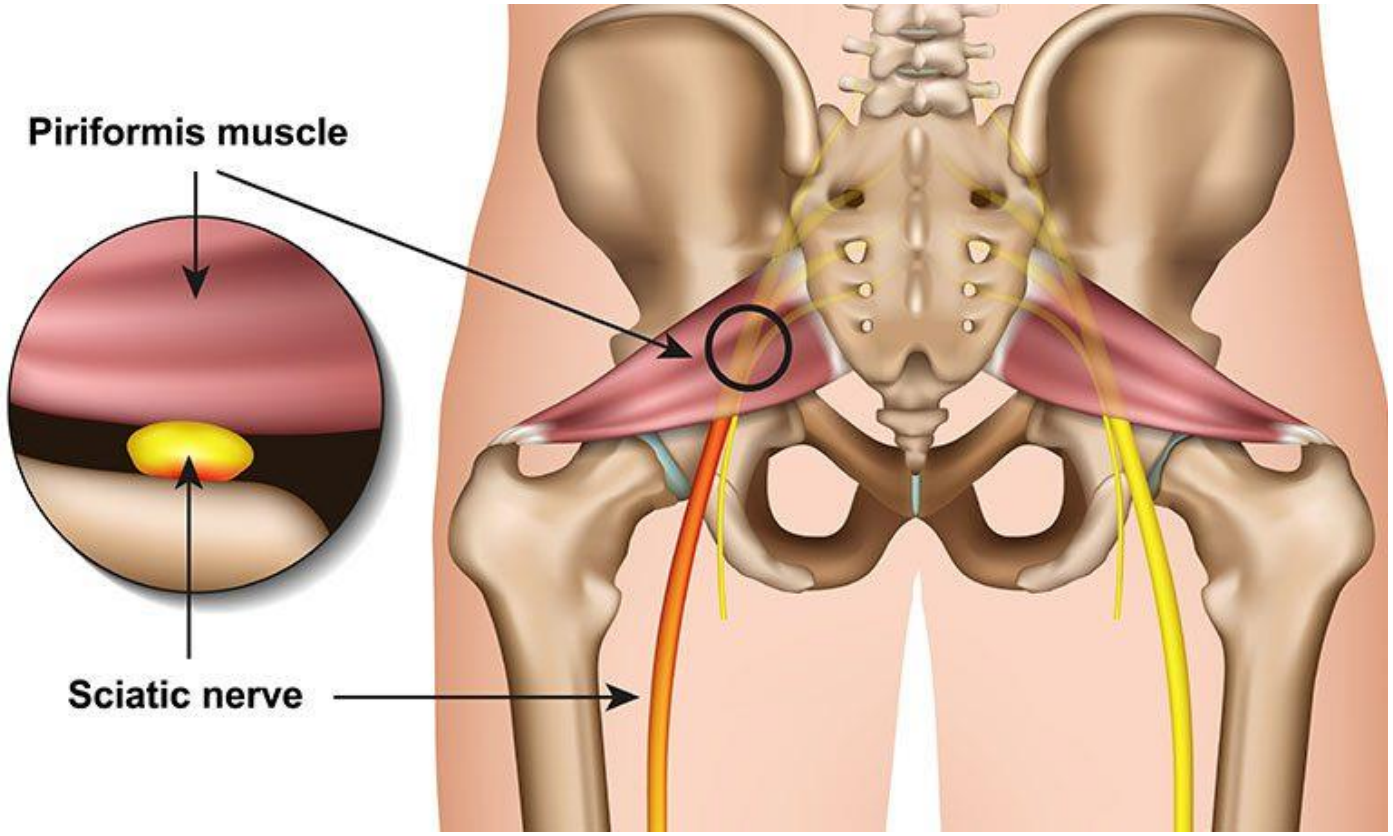
1- Piriformis syndrome

2- Gluteus max trigger points

3- Hamstring Syndrome

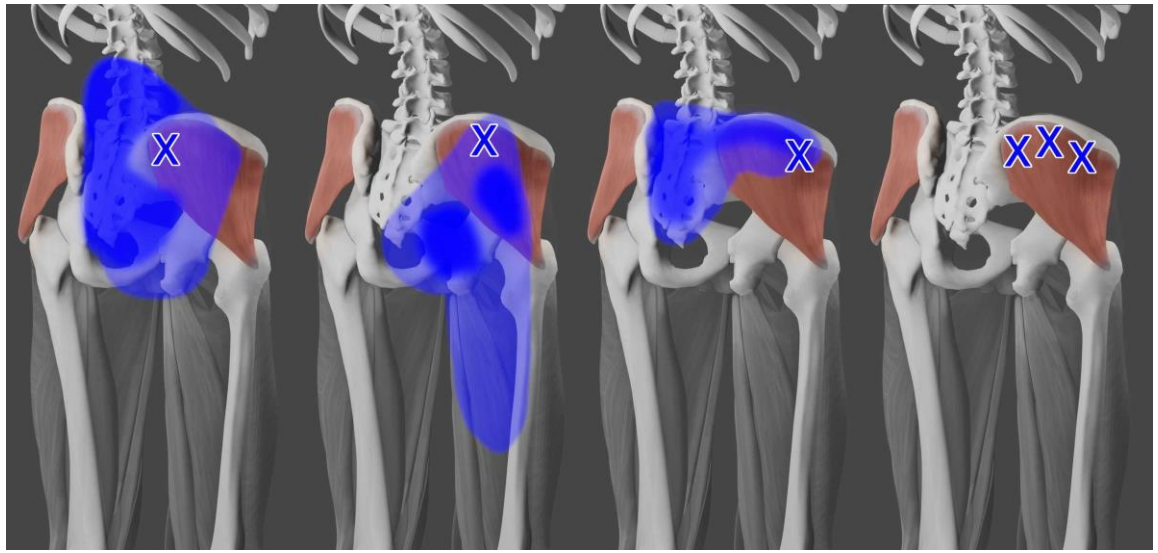
1) Piriformis Syndrome

- Tight piriformis muscle compresses the sciatic nerve in the deep gluteal region.
- Deep buttock pain → sometimes radiates to hamstring or calf
- Worse with sitting, climbing stairs, or hip rotation
- Tenderness on palpation over piriformis
- No numbness or true weakness
- Tests: FAIR test, Pace, palpation



2) Gluteus Medius / Minimus Trigger Points

- Trigger points refer pain along the lateral hip → outer thigh (can mimic L5 sciatica).
- Pain worse when lying on the affected side
- Painful to palpate over greater trochanter or gluteal belly
- Normal neurological exam



3) Hamstring Syndrome

- The sciatic nerve can be irritated at the proximal hamstring origin (ischial tuberosity).
- Pain local to ischial tuberosity
- Pain sitting on hard surfaces and with hamstring stretch
- Puranen-Orava Test → pain with stretching hamstrings in standing.